

CS-663

Digital Image Processing (DIP)

Basic Operations

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Digital Image

- Image representation: an image is a multi-dimensional array
- Array-element values can be represented as integers or floating-point
 - e.g., int8, uint8, int16, int32, float/single, double
- Image file format = **matrix** + **header**
- **Header** contains information like:
 - File format “signature” (specific sequence of first few bytes in file)
 - Array dimensions and size
 - Pixel size units in physical dimensions,
e.g., 1 x 1 x 1.5 mm (MRI image) or km (google maps)
 - Binary-data indicator (yes/no)
 - Binary-data-format indicator (big/little endian)

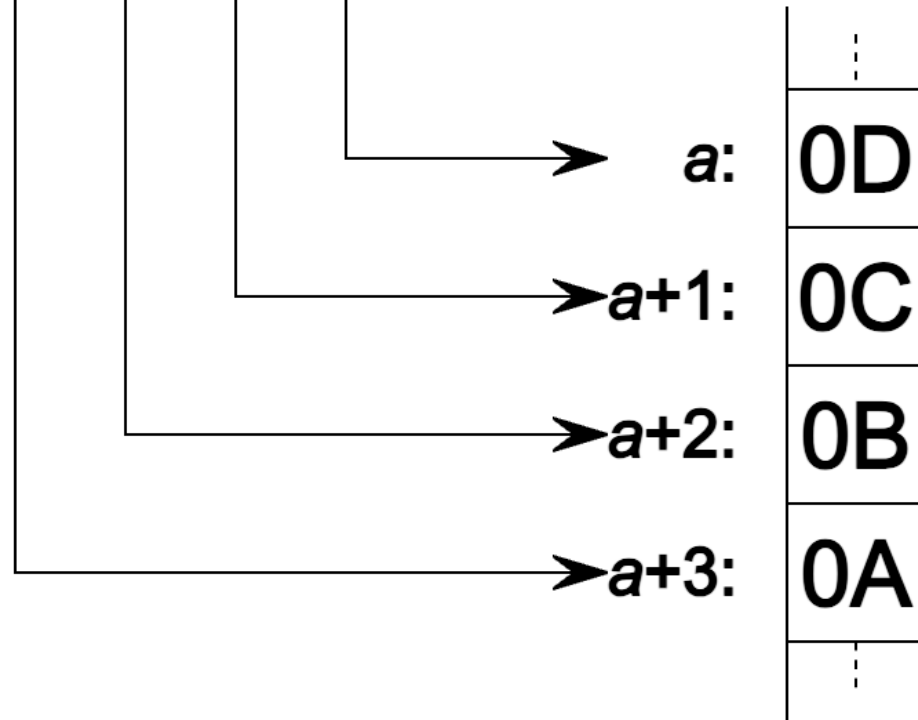
Digital Image

- Endianness: Big-endian format stores most-significant-byte (MSB) at lowest memory address

one 32-bit integer

0A0B0C0D

arranged as
four bytes in
memory

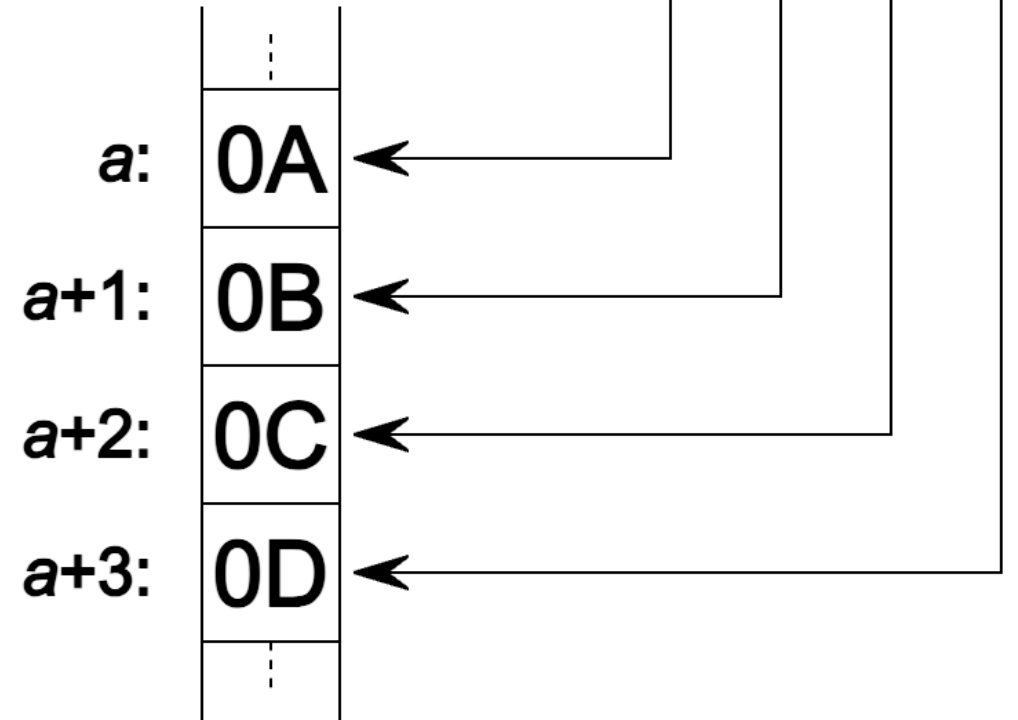


Little-endian

arranged as
four bytes in
memory

one 32-bit integer

0A0B0C0D



Big-endian

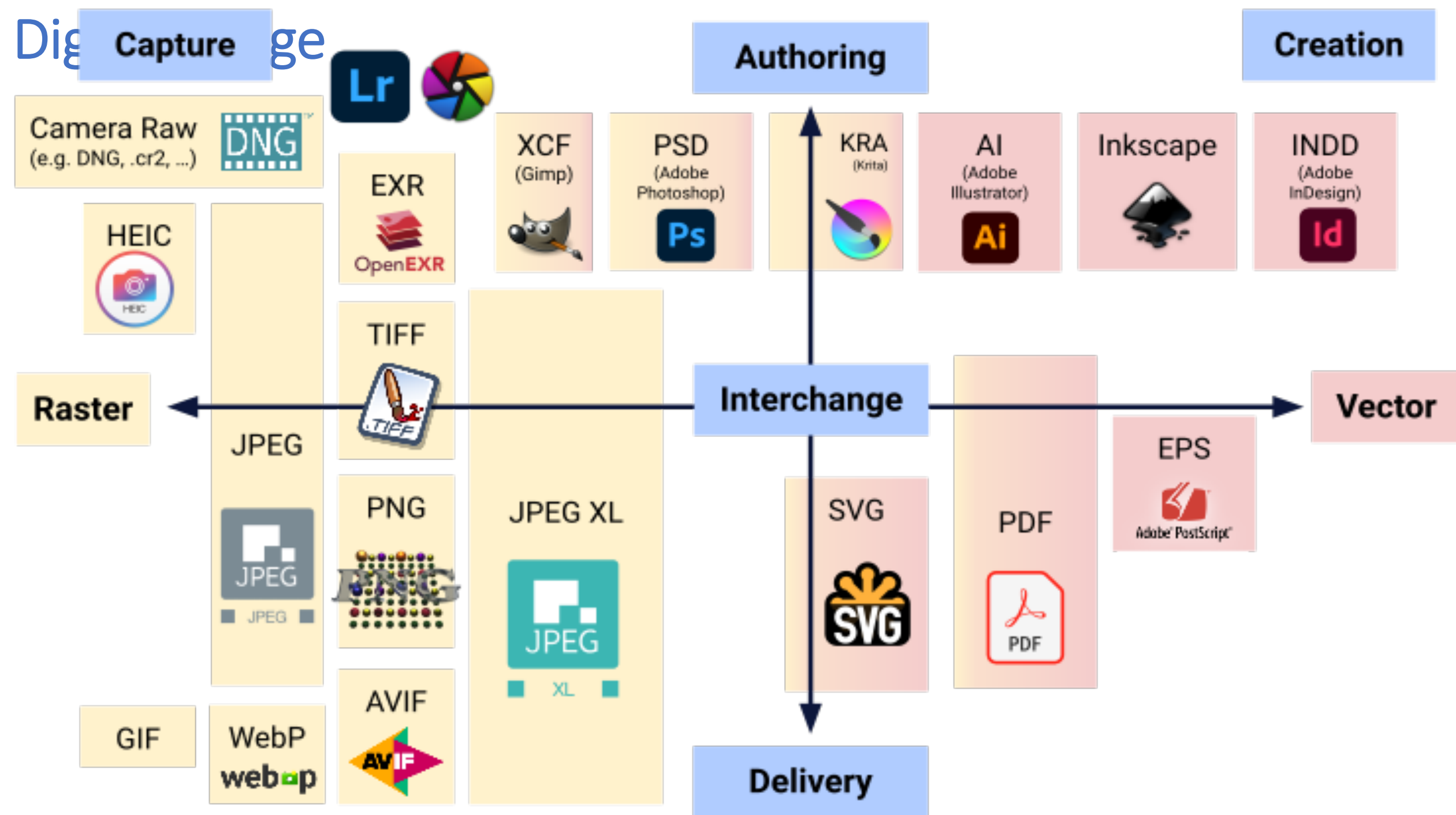
Digital Image

- P2 = “magic number”
 - Identifies file type
 - Pgm format stores numbers in ASCII
- # → comment
- 17 columns
- 7 rows
- 255 = maximum value

```
P2
# oie.pgm
17 7
255
0 13 13 13 13 13 13 13 0 0 0 0 0 0 0 0 0
0 13 0 0 0 0 0 13 0 7 7 0 0 81 81 81 81
0 13 0 7 7 7 0 13 0 7 7 0 0 81 0 0 0
0 13 0 7 0 7 0 13 0 7 7 0 0 81 81 81 0
0 13 0 7 7 7 0 13 0 7 7 0 0 81 0 0 0
0 13 0 0 0 0 0 13 0 7 7 0 0 81 81 81 81
0 13 13 13 13 13 13 13 0 0 0 0 0 0 0 0 0
```

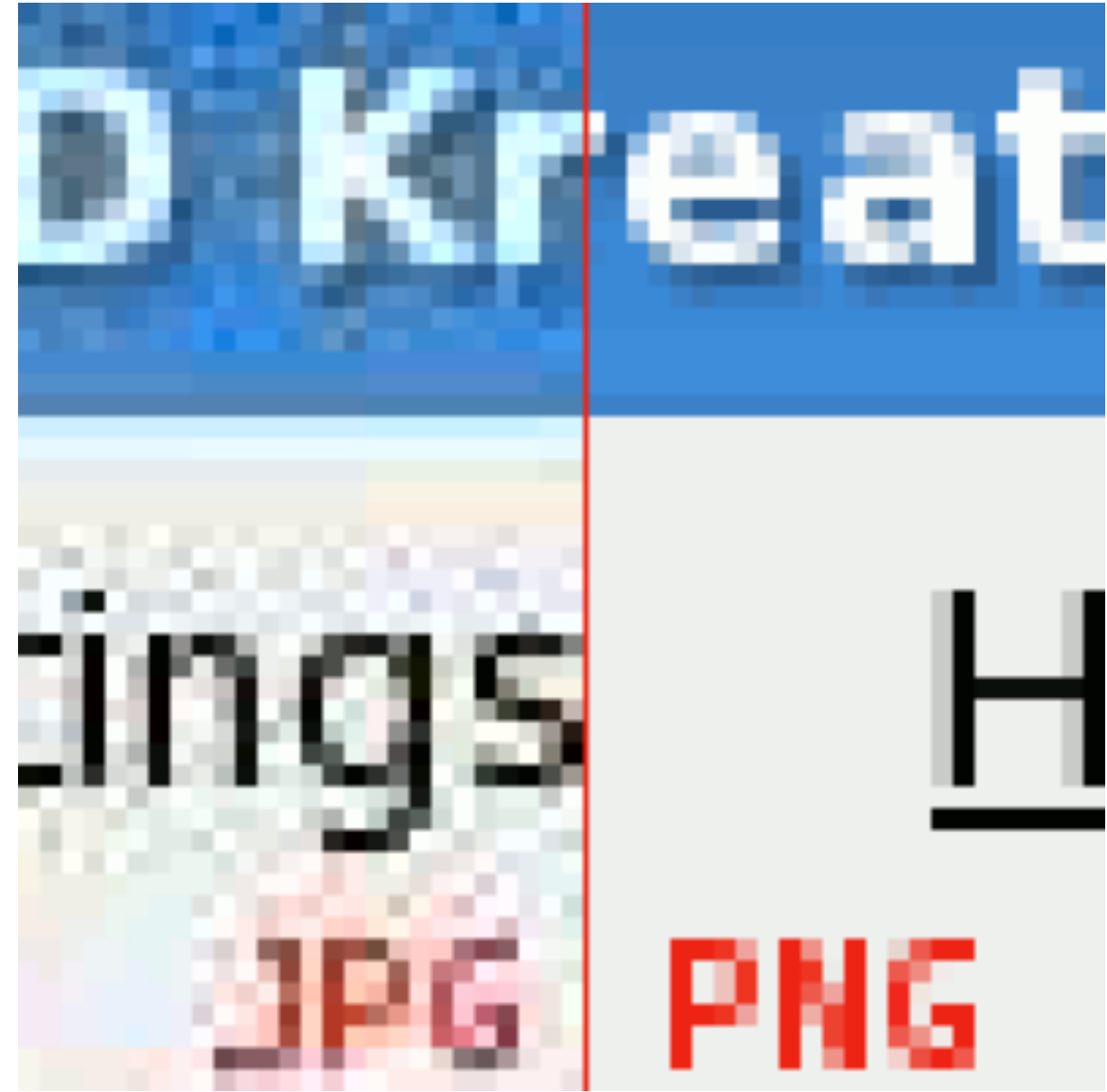


Digital Image



Digital Image

- Some image file formats use compression that can lose information
 - e.g., jpeg/jpg, jp2, mpeg, mp4, avi
- **Lossless**/NO compression is great for:
 - Photography: sports, fine art, nature
 - Scientific imaging
 - **All your assignments and course projects (unless otherwise stated)**
 - Format examples: **png, tiff**
OR **npz, mat**



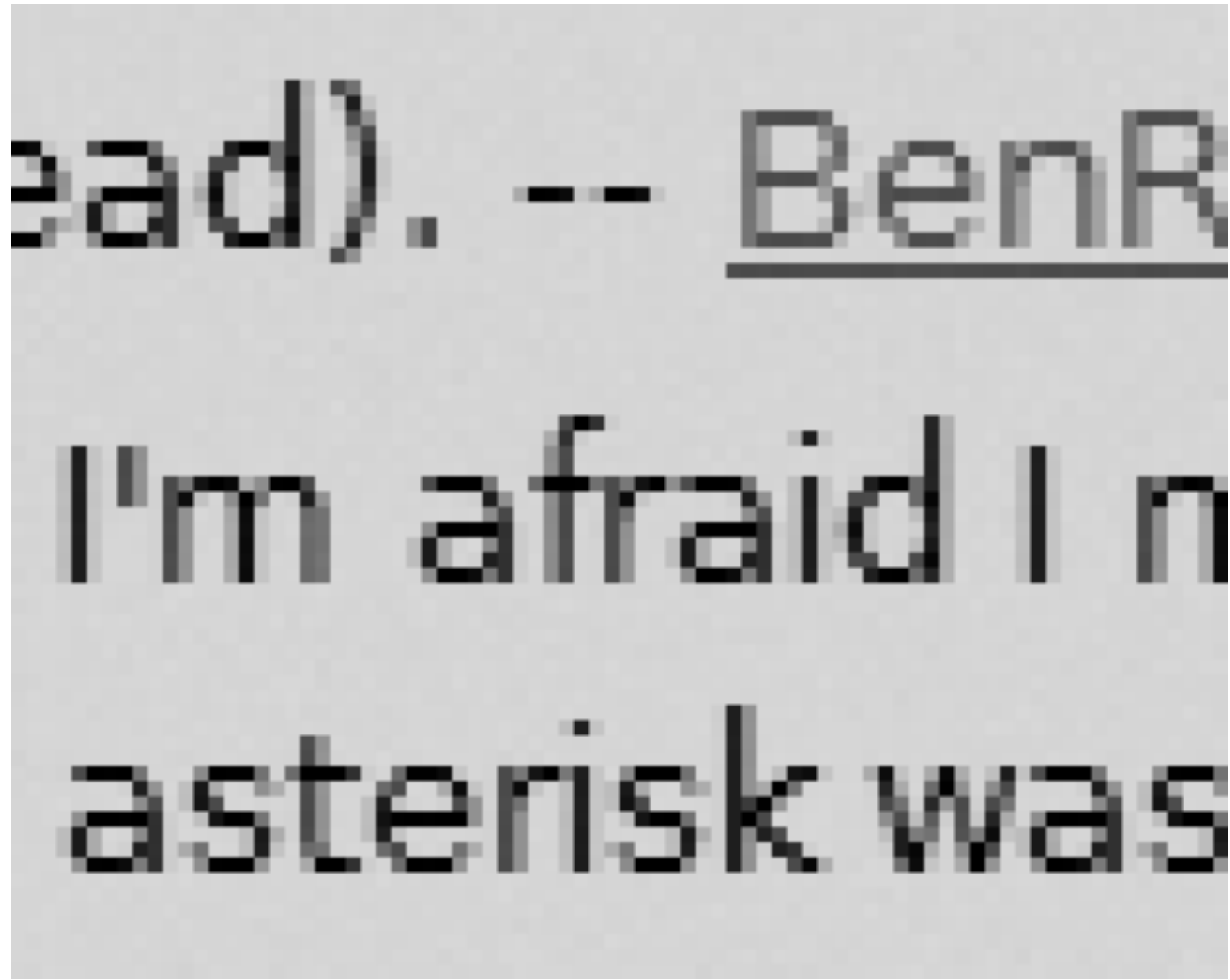
Digital Image

- JPEG Quality:
10, 1,
100, 50, 25
- Even quality 100 isn't
guaranteed lossless



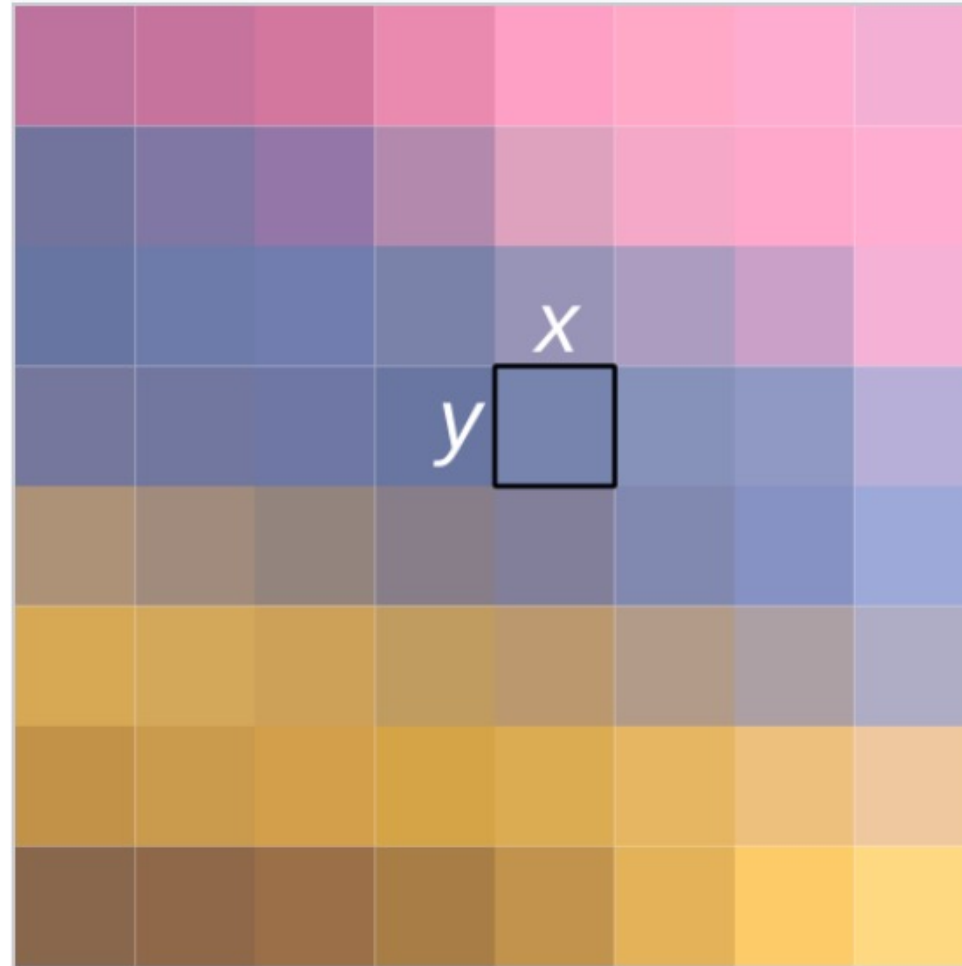
Digital Image

- Imperfections resulting from JPEG quality 90

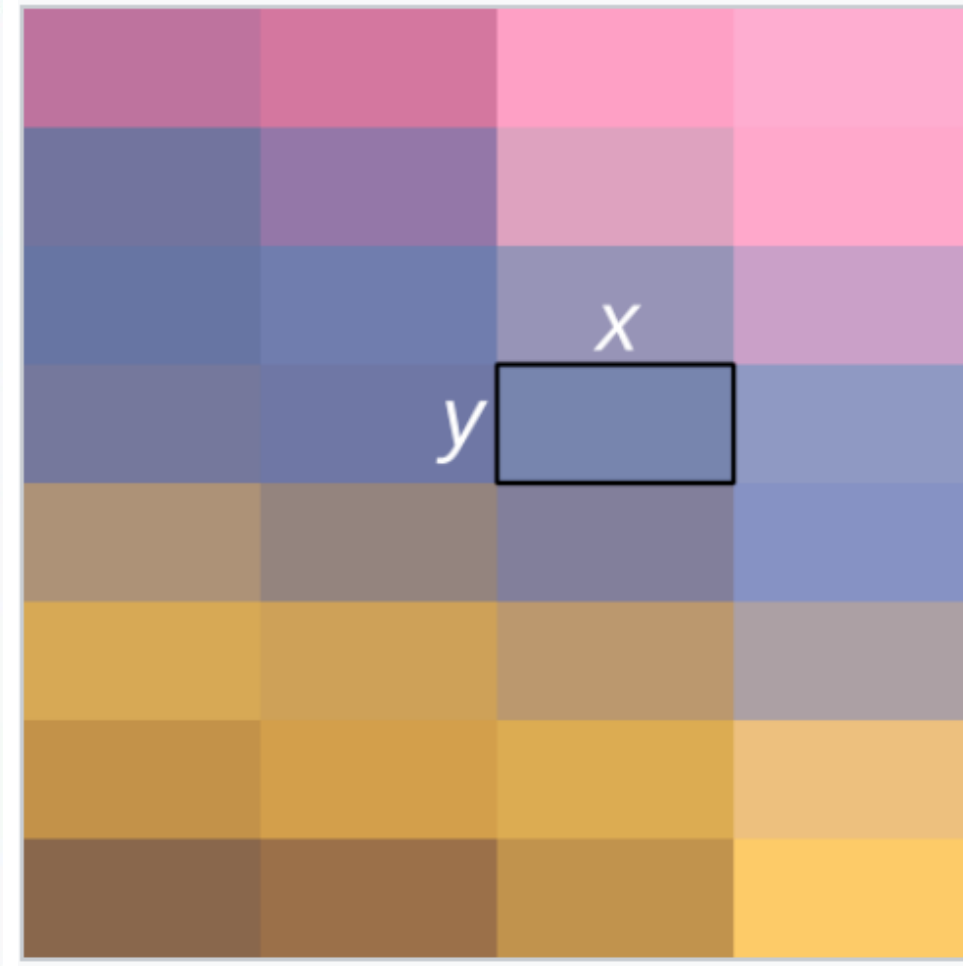


Digital Image

- [Pixel aspect ratio](#)
- Pixel size related to physical units: e.g., mm, km
- It corresponds to size of part of physical scene it represents
- **Pixels aren't always square**



Pixel aspect ratio 1:1



Pixel aspect ratio 2:1



Digital Image

- Satellite image
- Physical scale



Digital Image

- Image of same scene (as before) with half the number of pixels along left-right direction
- Effectively, pixel units (scale) are different along left-right and top-down directions
- But display doesn't respect that



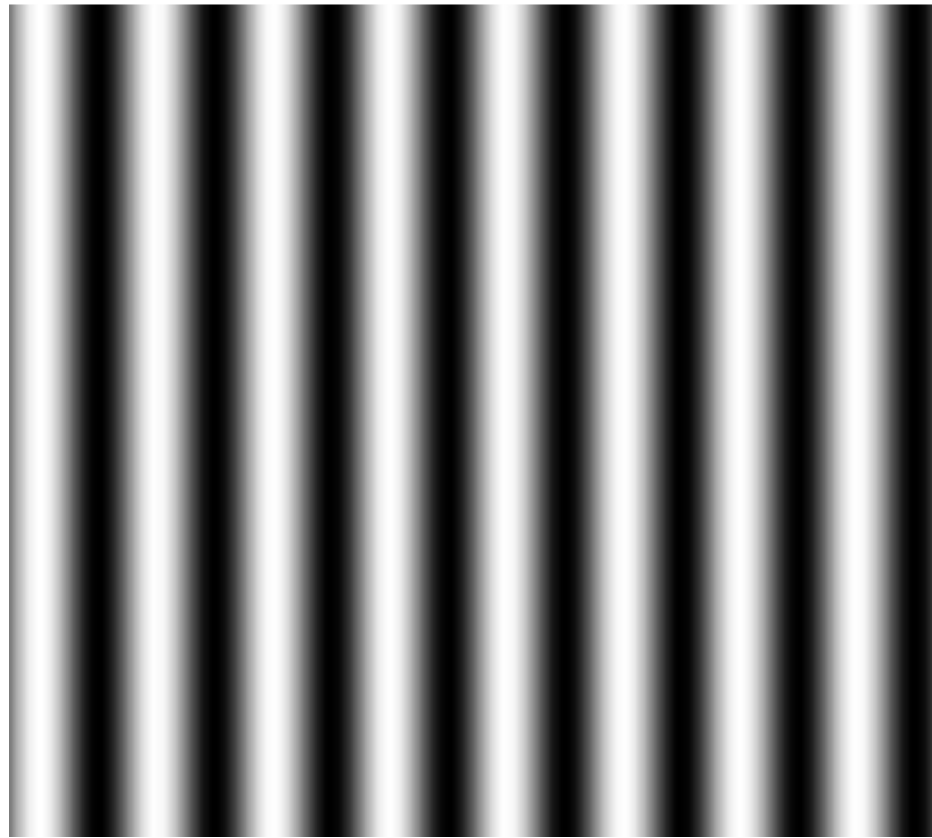
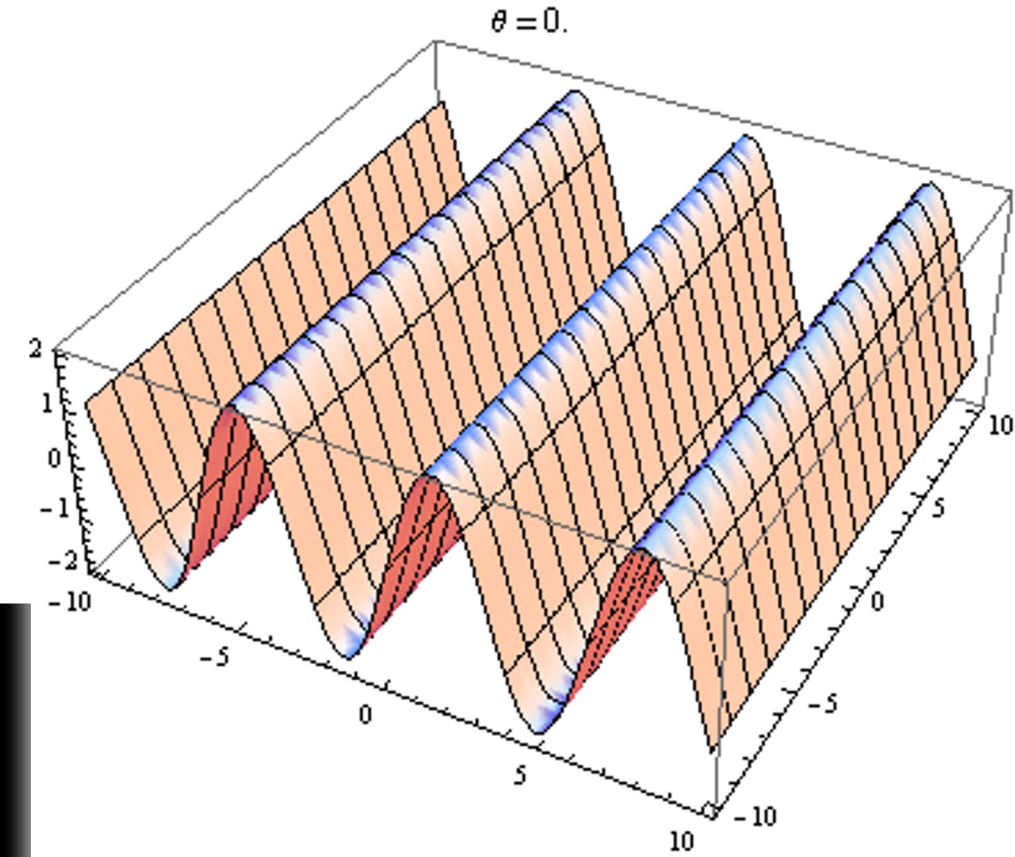
Digital Image

- Resizing bug (during editing)



Digital Image

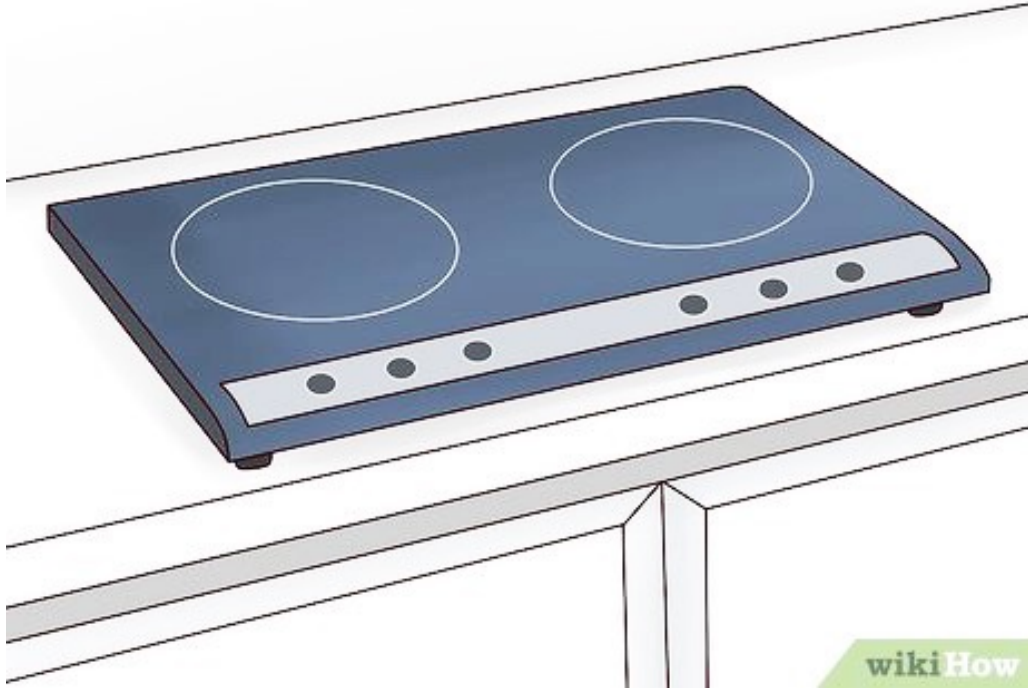
- Image can be modeled as a function
 - e.g., domain: (x,y) in $\mathbb{R}^2$
 - e.g., range: $f(x,y)$ in $\mathbb{R}$
- Sinusoid function



- Sinusoid image

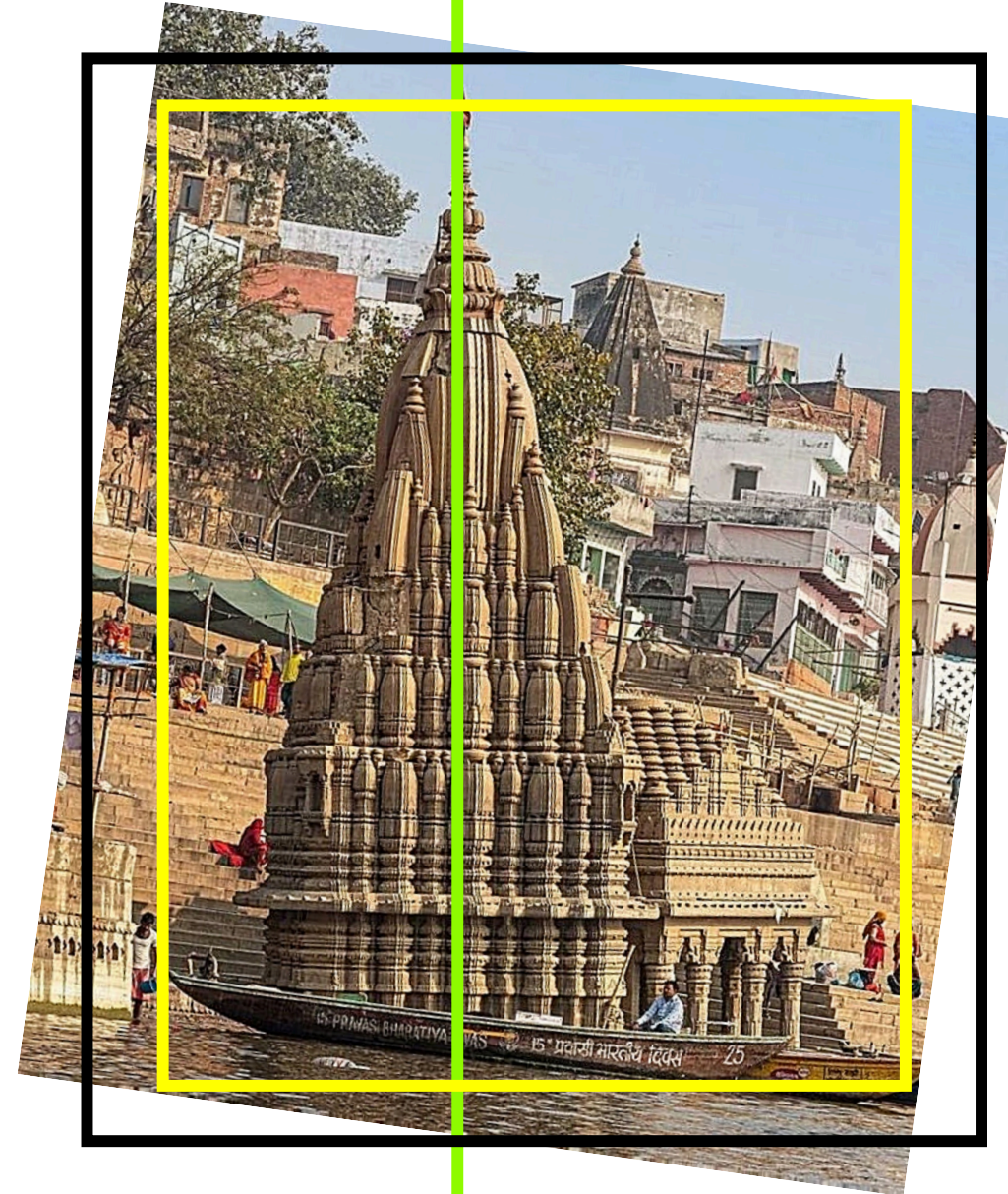
Digital Image

- Consider function $f(x,y,t)$ modeling temperature at time t , at location (x,y) on surface of a stove-top burner
- $f(x,y,t)$ visualized as a video →
- Colormap 



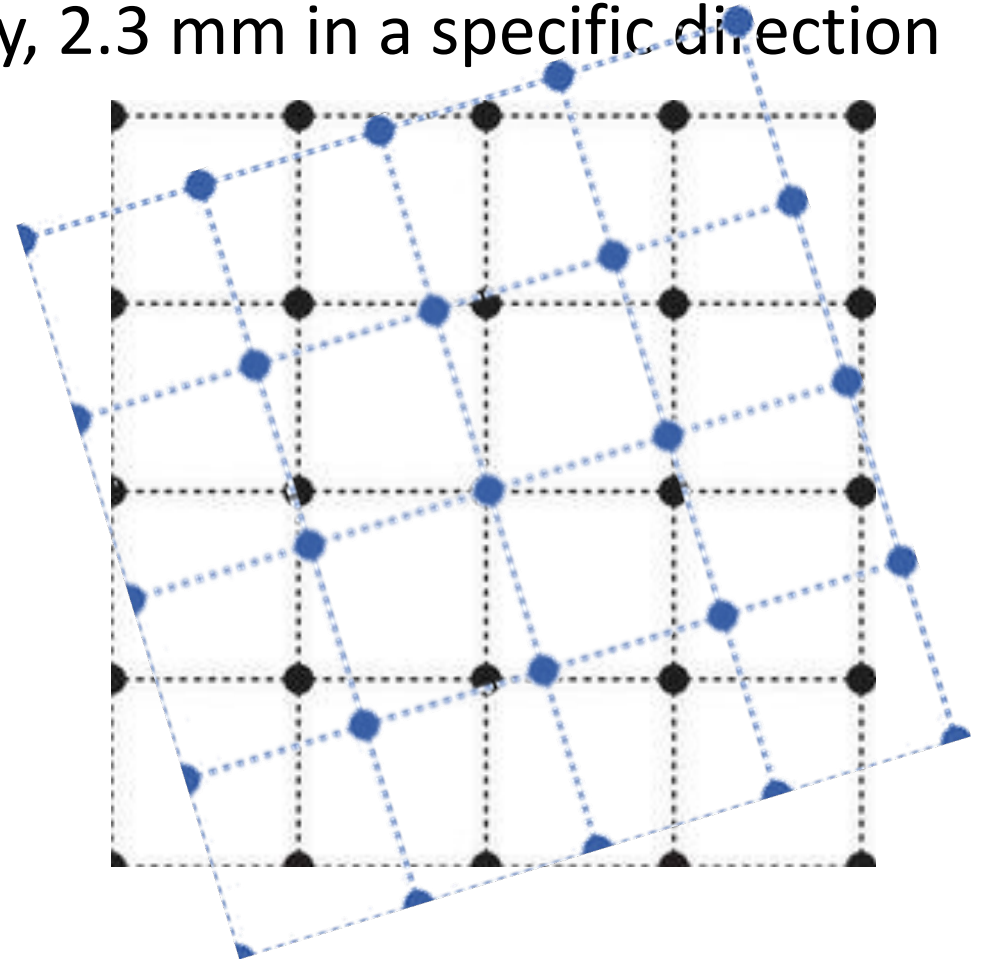
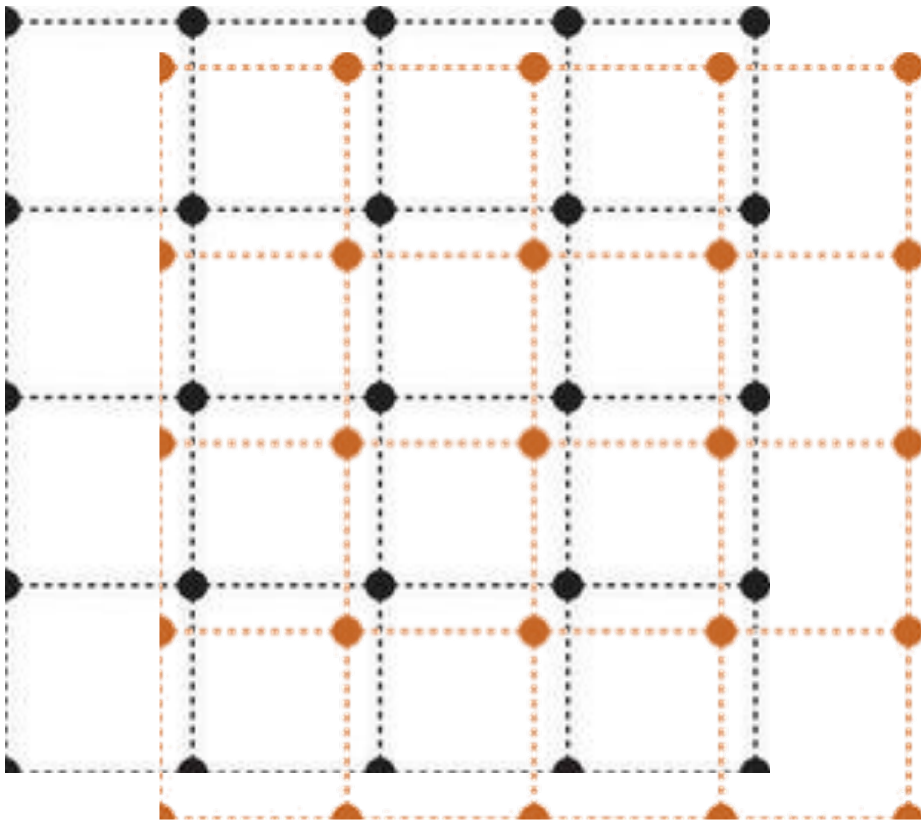
Digital Image Interpolation

- “Image rotation”
 - Rotating all objects in image
- Ratneshwar Mahadev temple
 - 9° tilt
 - Garbhagriha submerged under Ganga most of the year



Digital Image Interpolation

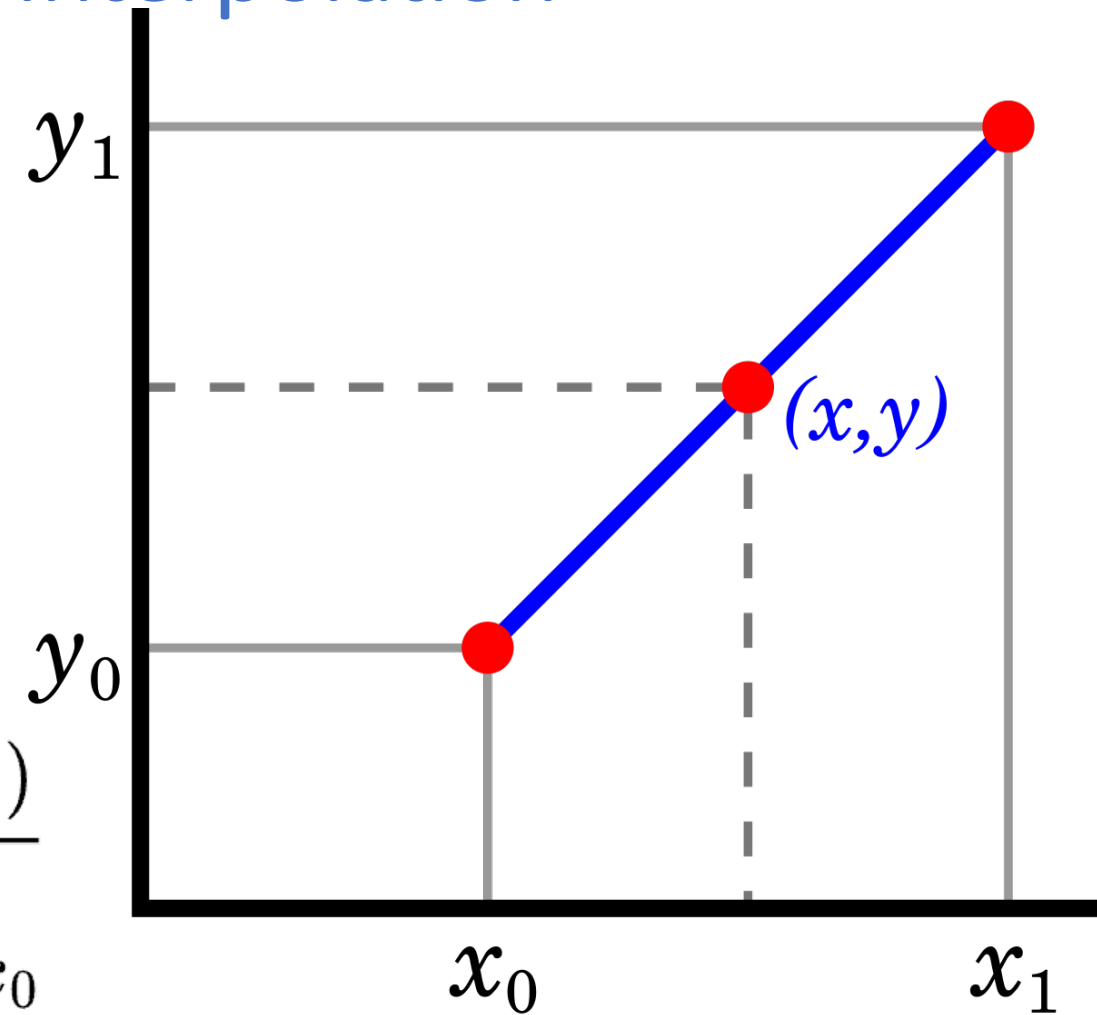
- Interpolating functions
- Motivation
 - Rotate (objects in) image by, say, 4.5 degrees, around a given point
 - Shift/translate (all objects in) image by, say, 2.3 mm in a specific direction



Digital Image Interpolation: Linear Interpolation

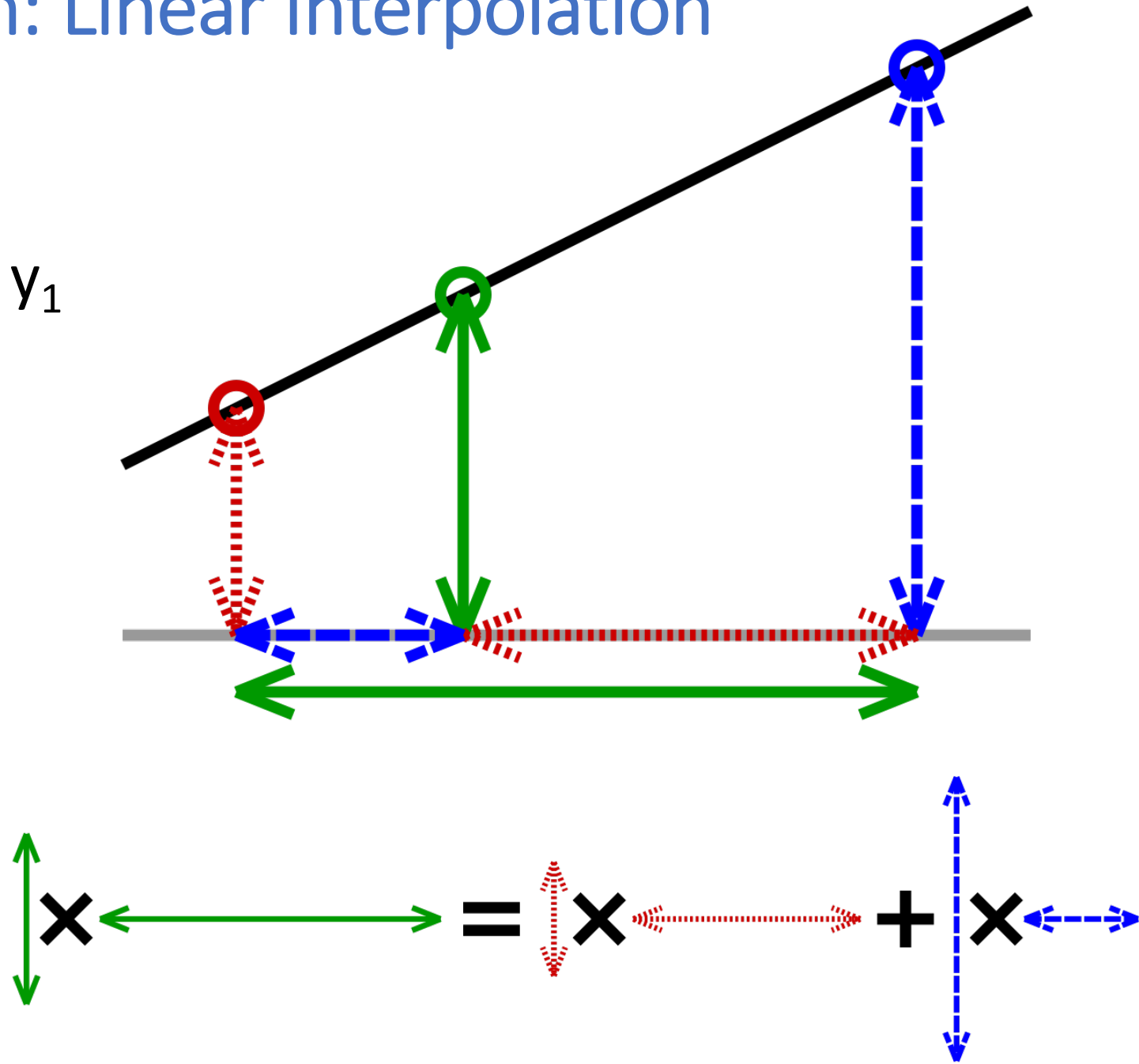
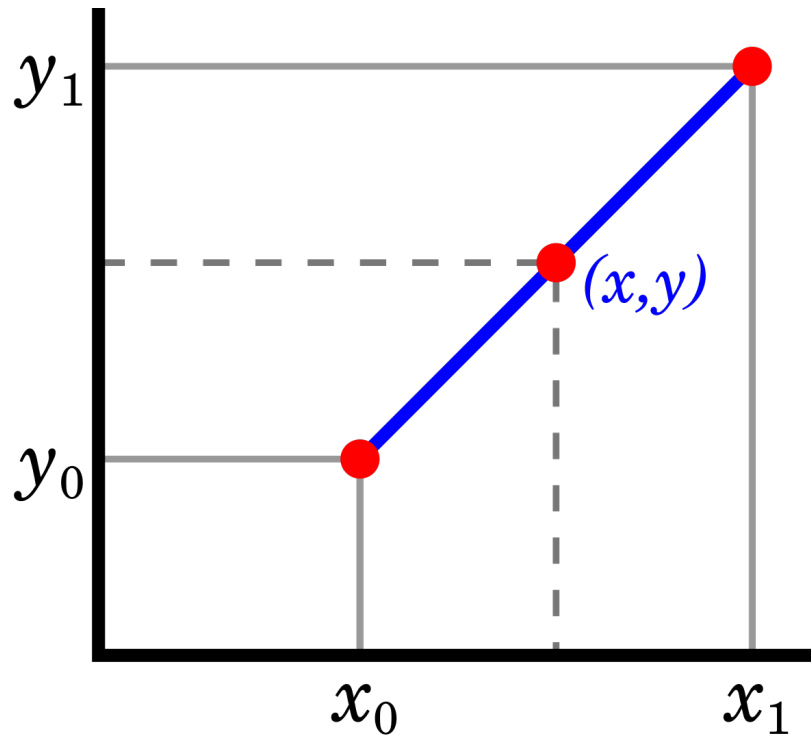
- $y = f(x)$ where x is scalar
- Given: (x_0, y_0) , (x_1, y_1)
- Given: x within interval $[x_0, x_1]$
- Estimate y

$$\begin{aligned} y &= y_0 + (x - x_0) \frac{y_1 - y_0}{x_1 - x_0} \\ &= \frac{y_0(x_1 - x_0)}{x_1 - x_0} + \frac{y_1(x - x_0) - y_0(x - x_0)}{x_1 - x_0} \\ &= \frac{y_1 x - y_1 x_0 - y_0 x + y_0 x_0 + y_0 x_1 - y_0 x_0}{x_1 - x_0} \\ &= \frac{y_0(x_1 - x) + y_1(x - x_0)}{x_1 - x_0}, \end{aligned}$$

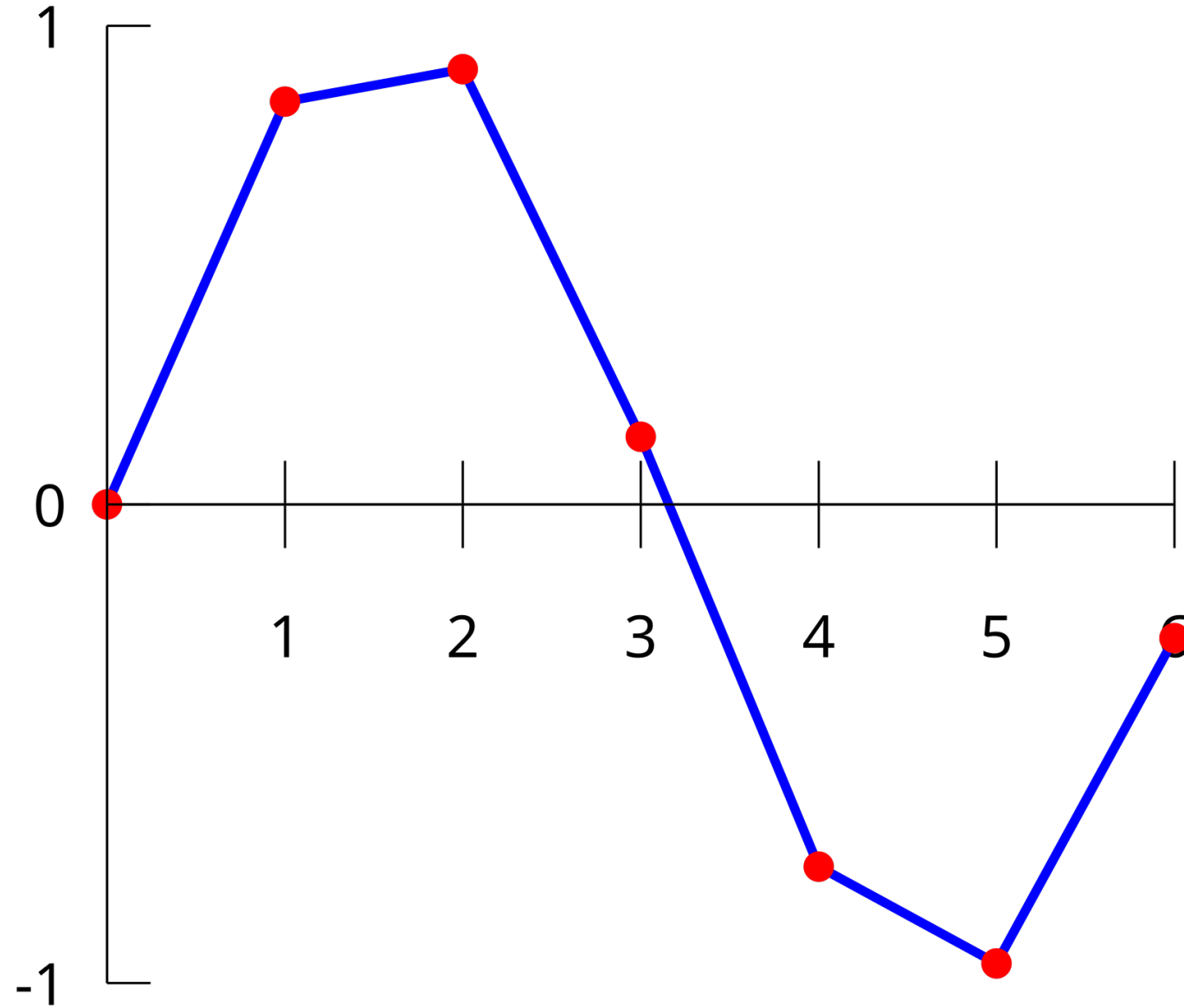


Digital Image Interpolation: Linear Interpolation

- $y = \frac{y_0(x_1 - x) + y_1(x - x_0)}{x_1 - x_0}$
- Convex combination of y_0 and y_1
- y always lies within $[y_0, y_1]$
- Geometric interpretation

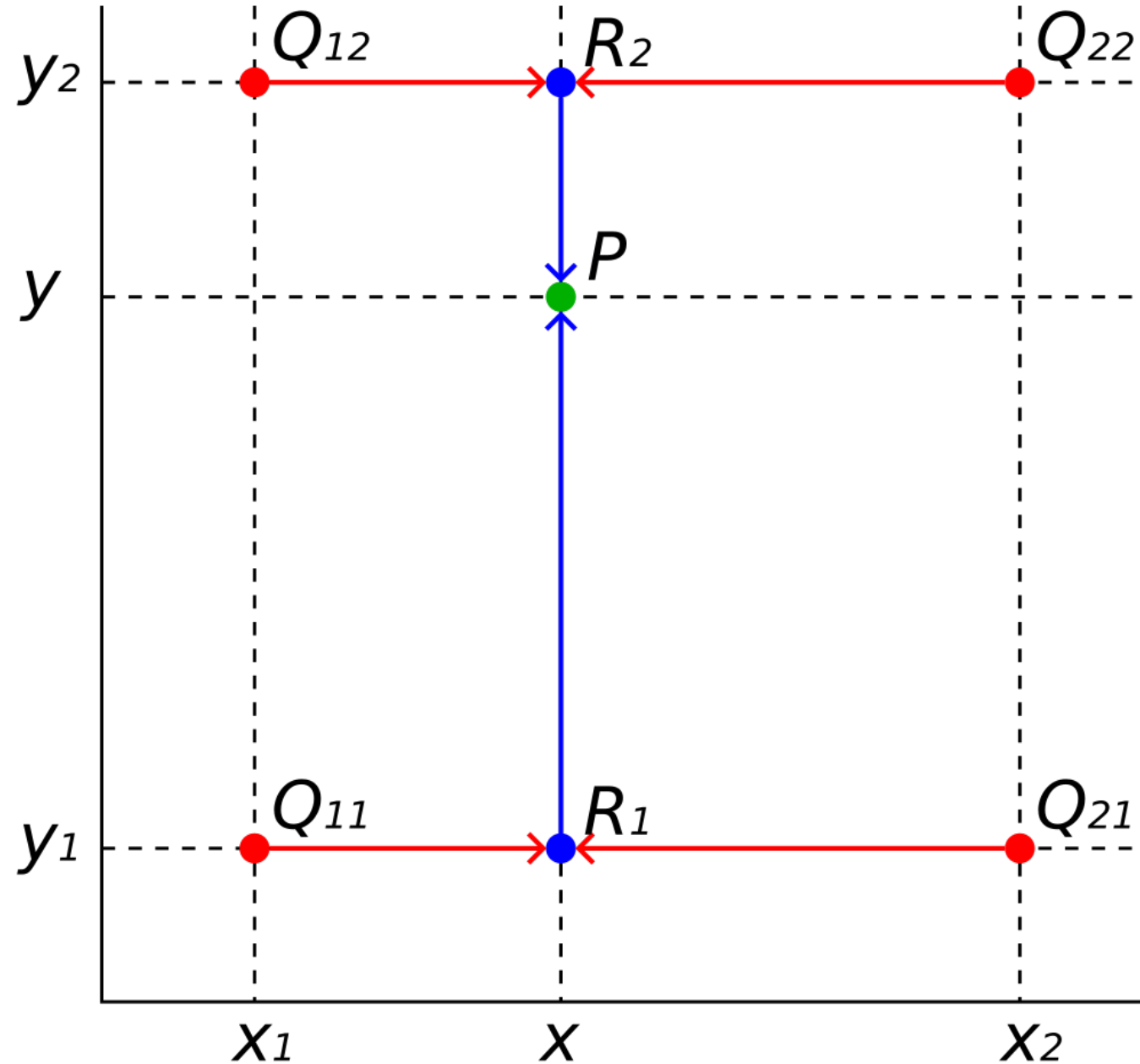


Digital Image Interpolation: Linear Interpolation



Digital Image Interpolation: Bilinear Interpolation

- $z = f(x,y)$
- Locations $Q_{ij} := (x_i, y_j)$
- Given:
 $(Q_{11}, f(Q_{11})), (Q_{12}, f(Q_{12})),$
 $(Q_{21}, f(Q_{21})), (Q_{22}, f(Q_{22}))$
- Given:
location $P := (x,y)$ where
 x within interval $[x_1, x_2]$,
 y within interval $[y_1, y_2]$
- Estimate:
value $f(P)$



Digital Image Interpolation: Bilinear Interpolation

- First, linearly interpolate along x axis

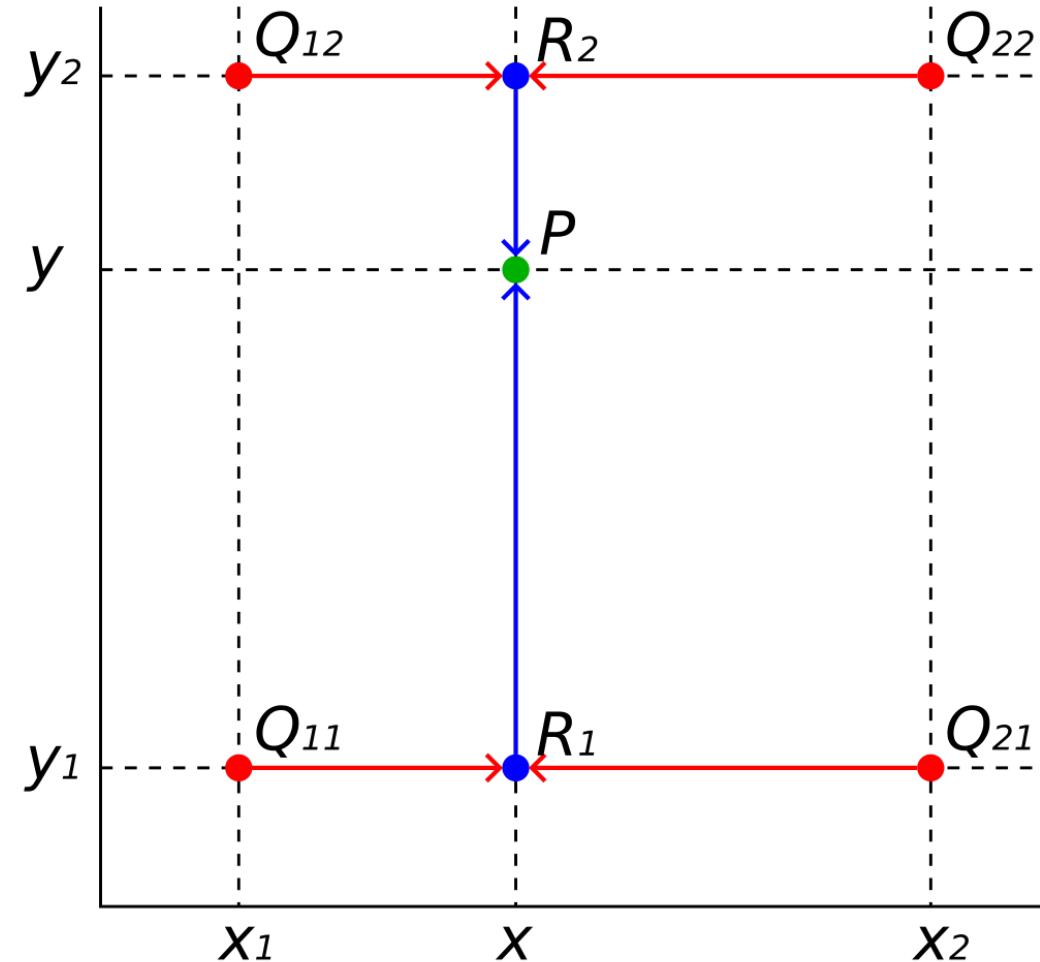
$$f(x, y_1) = \frac{x_2 - x}{x_2 - x_1} f(Q_{11}) + \frac{x - x_1}{x_2 - x_1} f(Q_{21})$$

$$f(x, y_2) = \frac{x_2 - x}{x_2 - x_1} f(Q_{12}) + \frac{x - x_1}{x_2 - x_1} f(Q_{22})$$

- Then, linearly interpolate along y axis

$$f(x, y) = \frac{y_2 - y}{y_2 - y_1} f(x, y_1) + \frac{y - y_1}{y_2 - y_1} f(x, y_2)$$

$$\begin{aligned} &= \frac{1}{(x_2 - x_1)(y_2 - y_1)} (f(Q_{11})(x_2 - x)(y_2 - y) + f(Q_{12})(x_2 - x)(y - y_1) \\ &\quad + f(Q_{21})(x - x_1)(y_2 - y) + f(Q_{22})(x - x_1)(y - y_1)) \end{aligned}$$

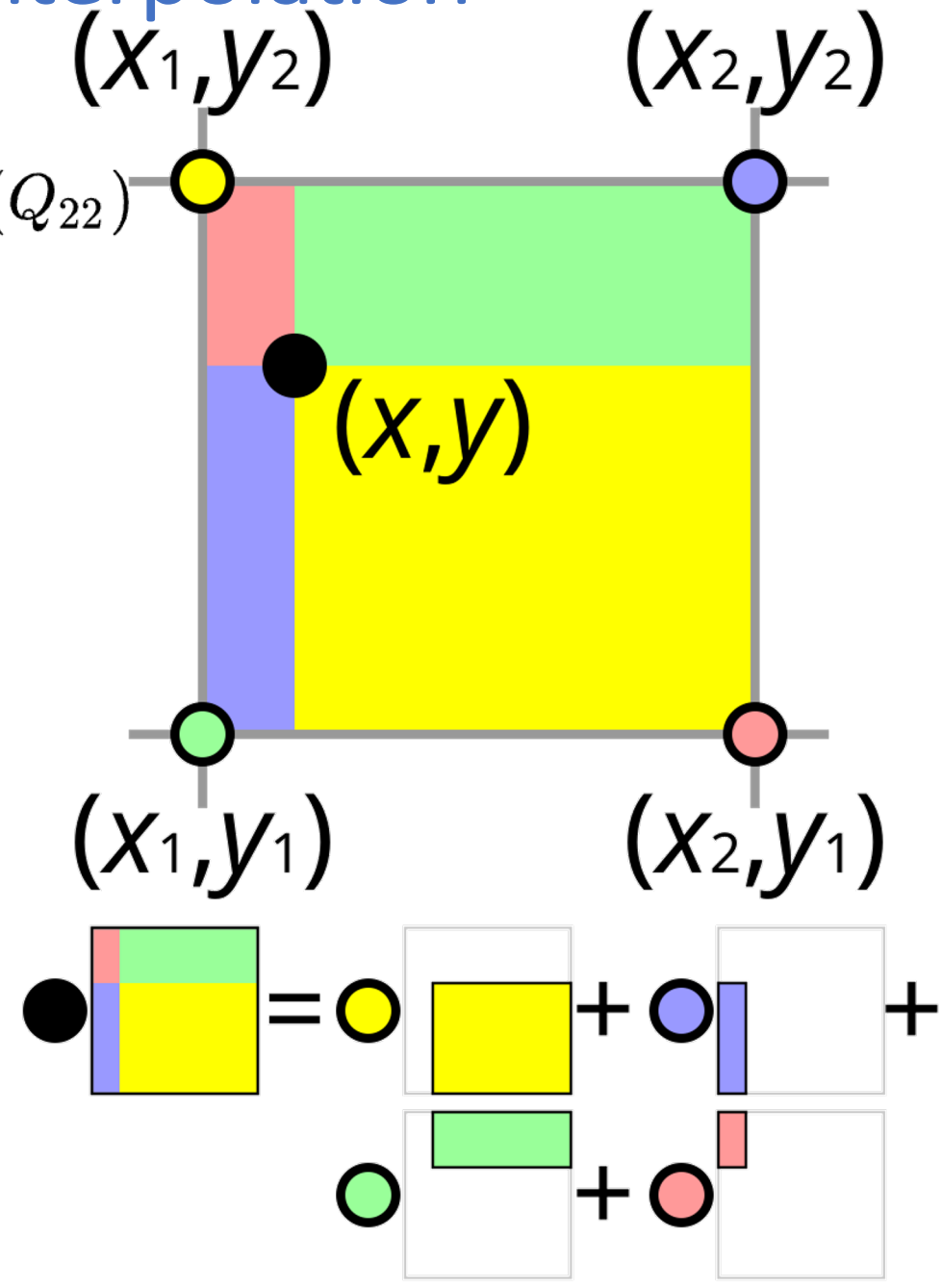


Digital Image Interpolation: Bilinear Interpolation

- So

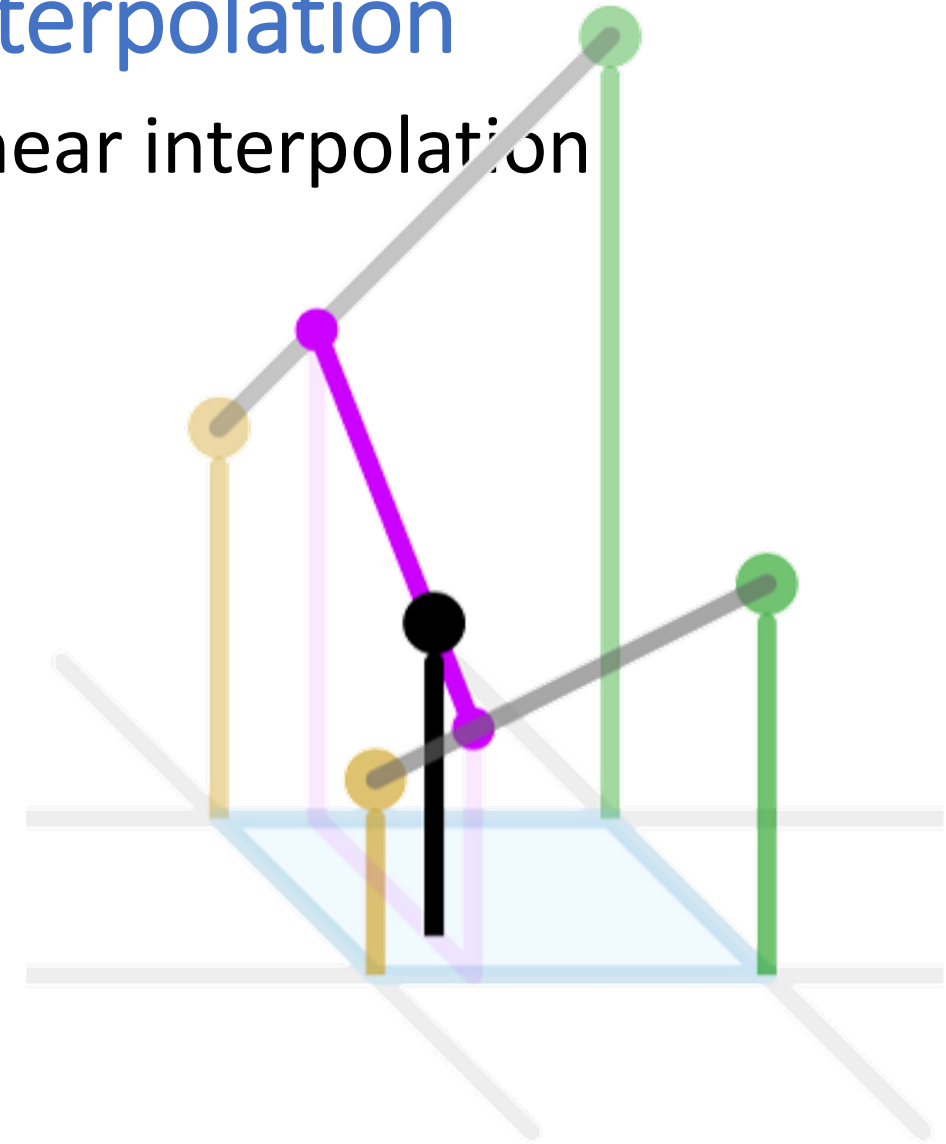
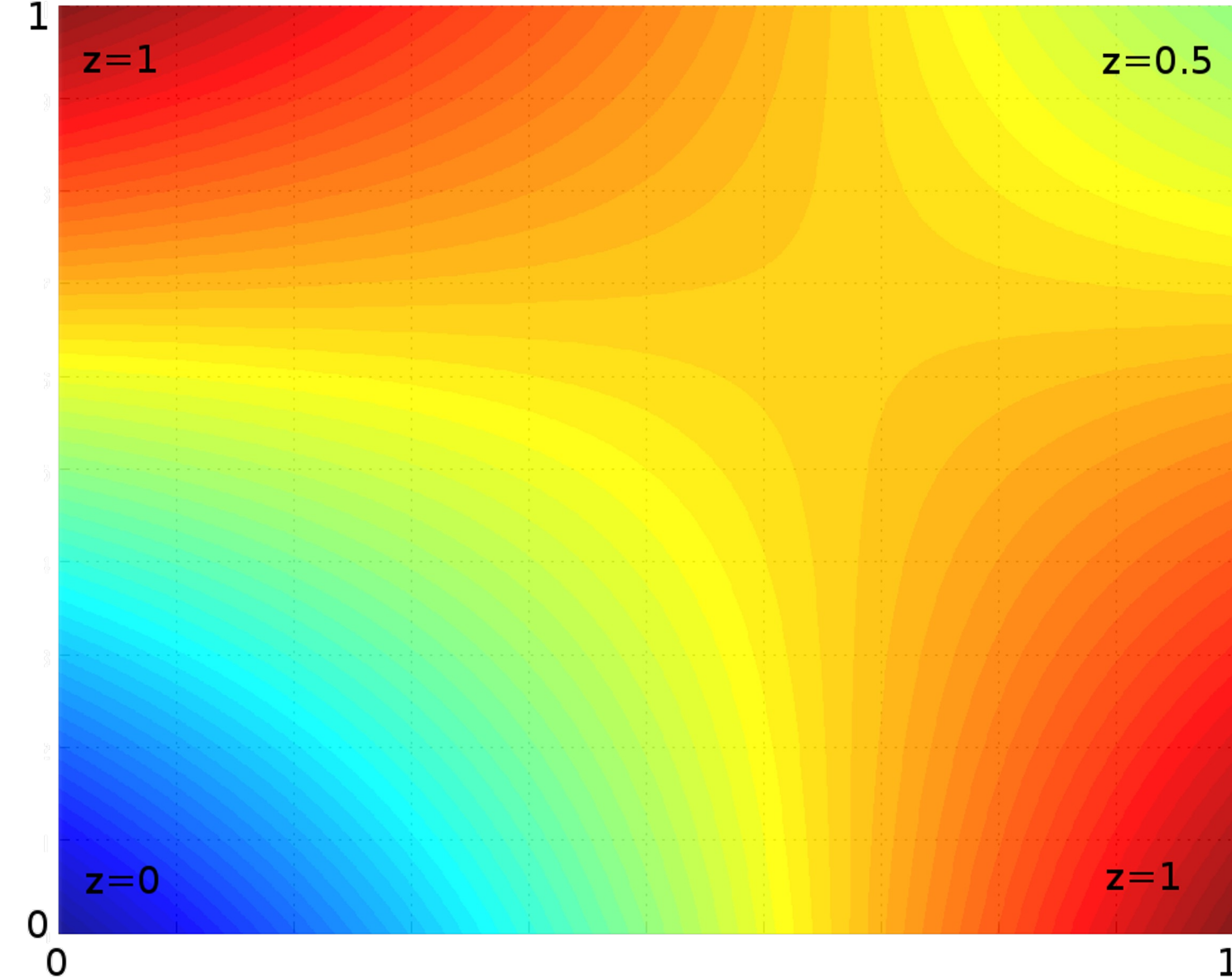
$$f(x, y) \approx w_{11} f(Q_{11}) + w_{12} f(Q_{12}) + w_{21} f(Q_{21}) + w_{22} f(Q_{22})$$

- which is a convex combination of $f(Q_{11})$, $f(Q_{12})$, $f(Q_{21})$, $f(Q_{22})$
- $f(x, y)$ guaranteed to lie within $(\min[f(Q_{11}), f(Q_{12}), f(Q_{21}), f(Q_{22})], \max[f(Q_{11}), f(Q_{12}), f(Q_{21}), f(Q_{22})])$
- Geometric interpretation
- Independent of order of interpolation steps along X and Y axes



Digital Image Interpolation: Bilinear Interpolation

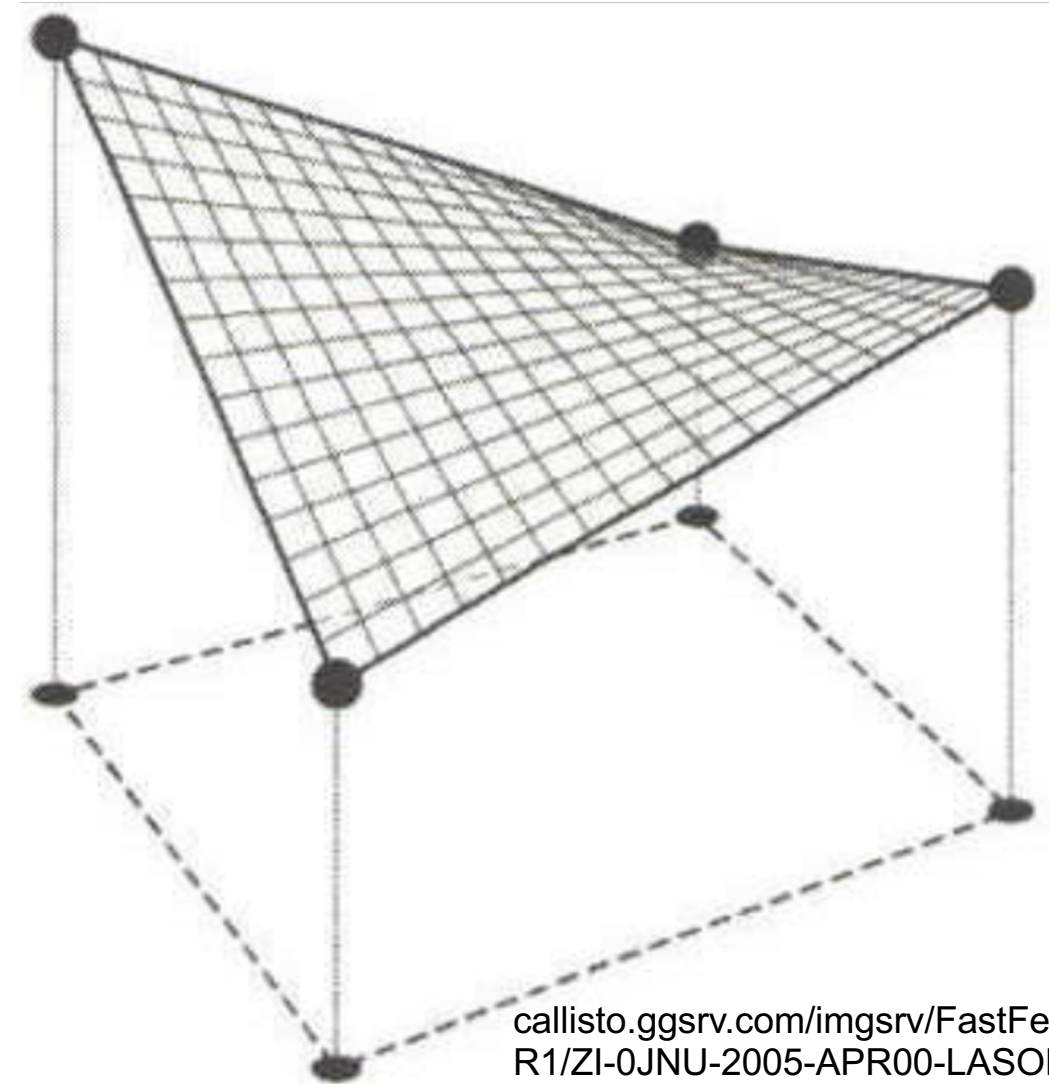
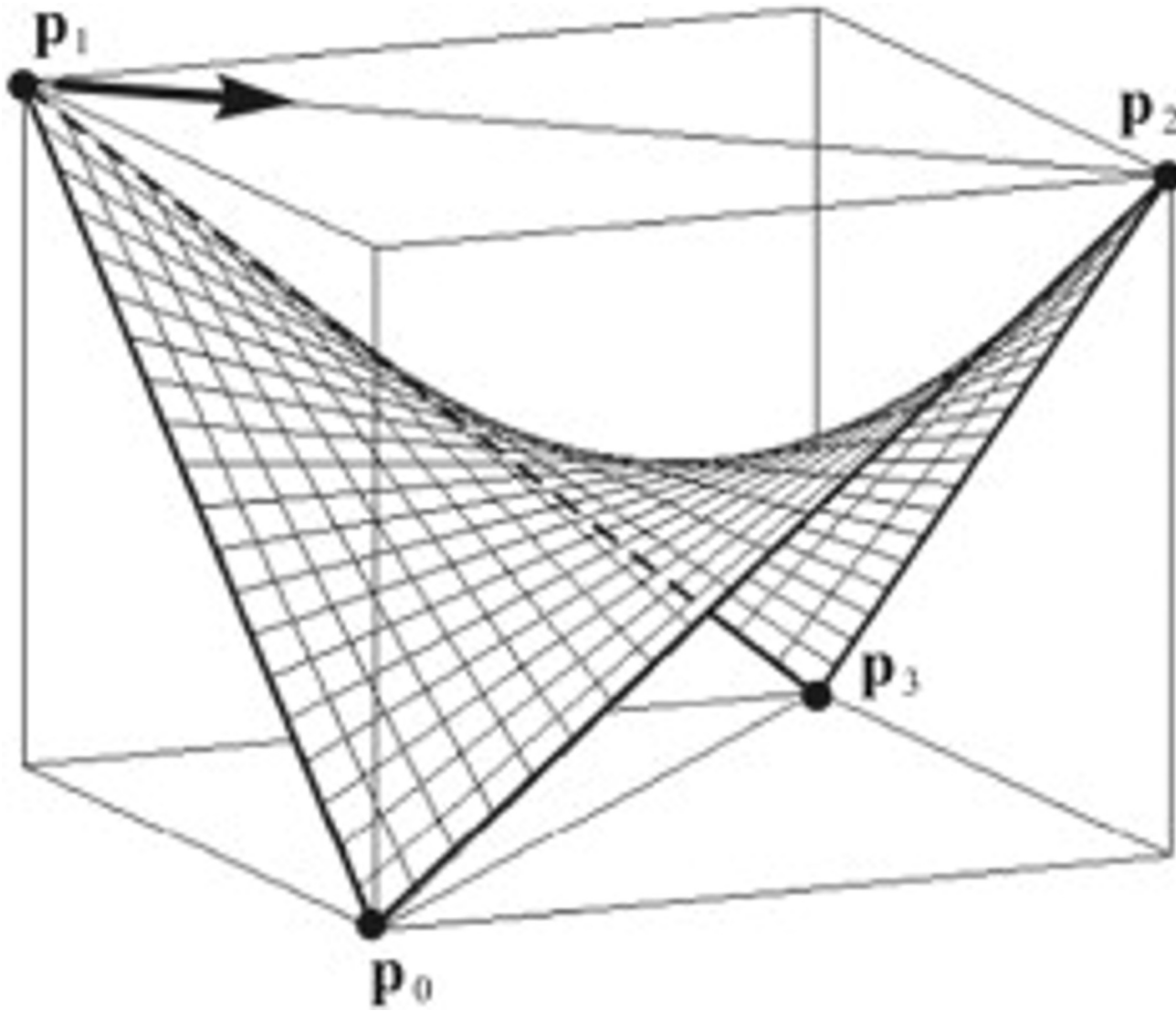
- Interpolated surface can be curved, unlike linear interpolation



Bilinear

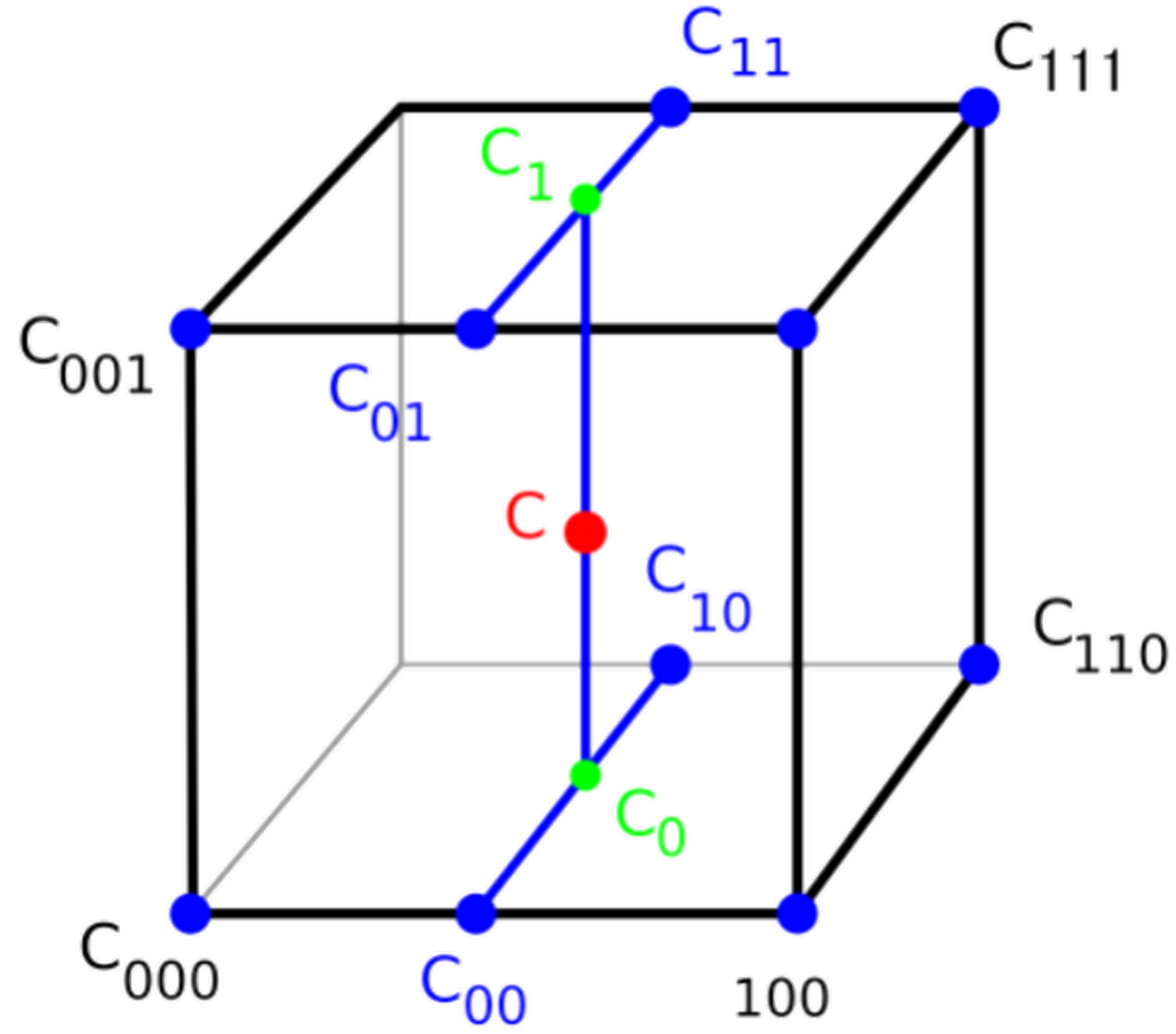
Digital Image Interpolation: Bilinear Interpolation

- Interpolated surface is curved, unlike linear interpolation



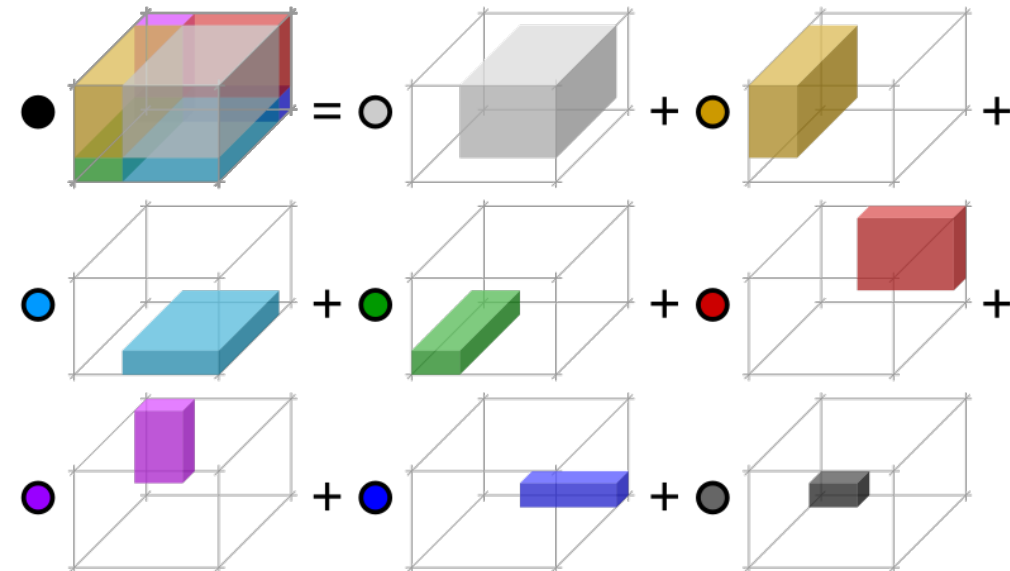
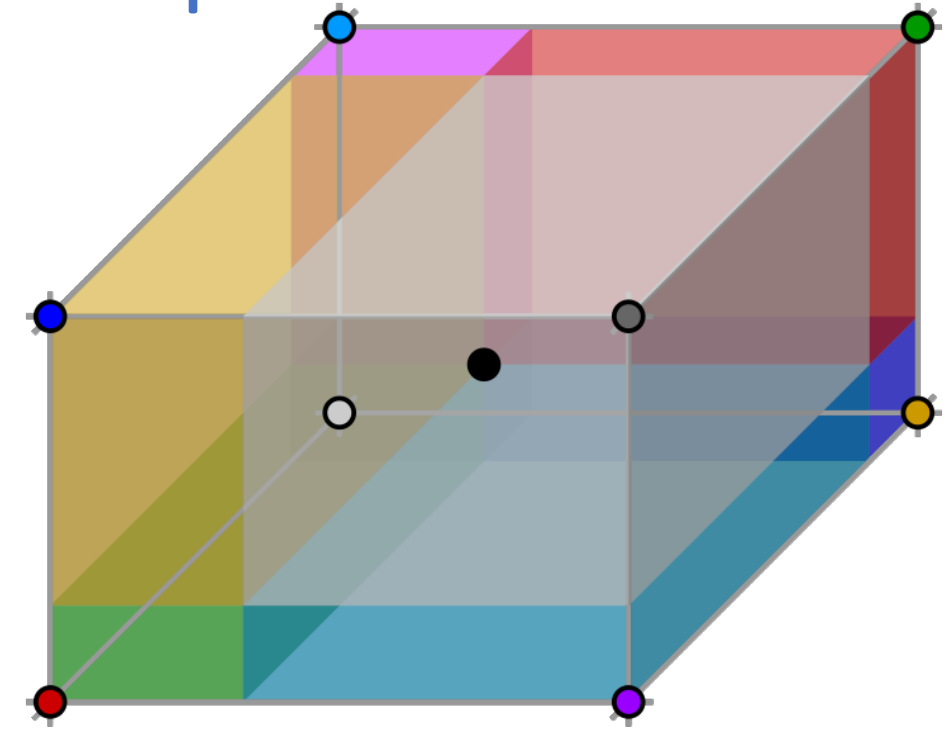
Digital Image Interpolation: Trilinear Interpolation

- Trilinear interpolation
- 2 bilinear interpolations followed by 1 linear interpolation
- Independent of order of interpolation steps along 3 axes



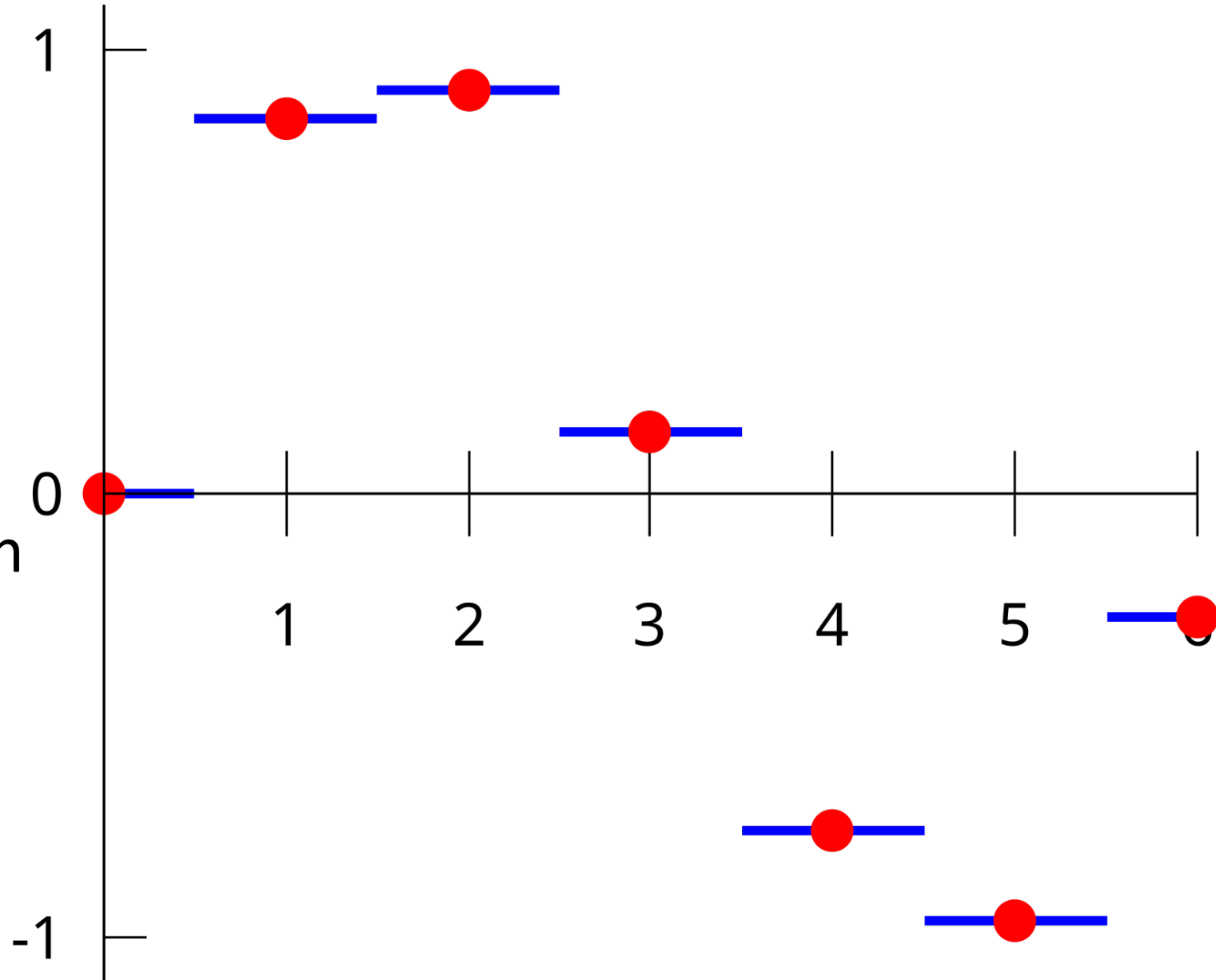
Digital Image Interpolation: Trilinear Interpolation

- Geometric interpretation
- Value at desired point *
entire volume
=
summation over all corners *
value at each corner *
partial volume diagonally opposite corner



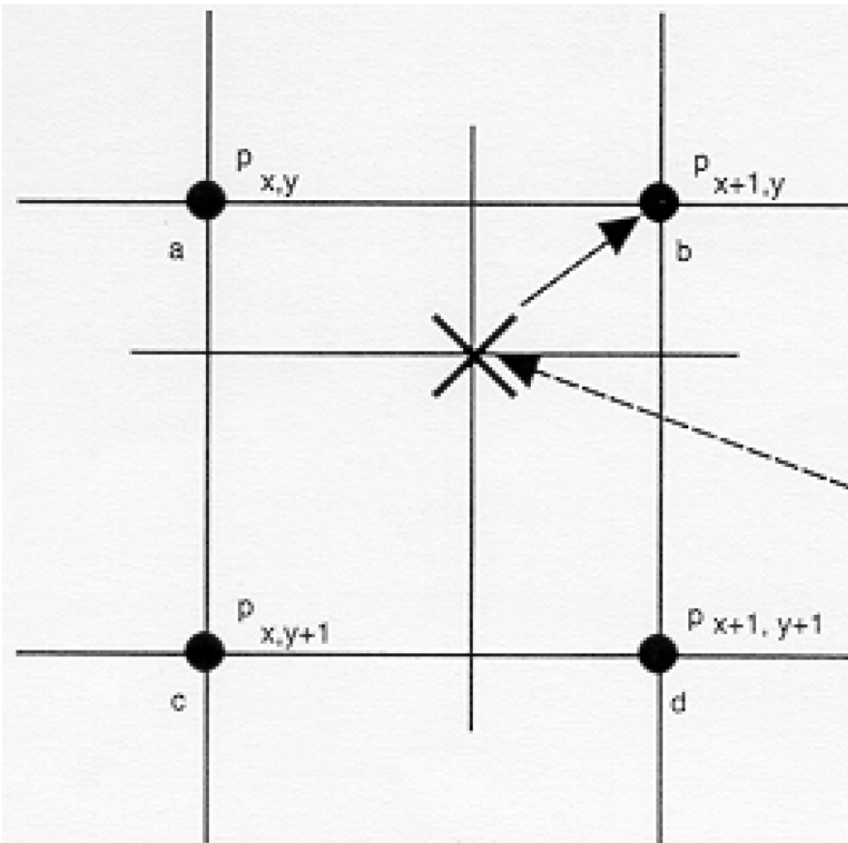
Digital Image Interpolation: Nearest-Neighbor

- For scalar variable x
- Given:
 $(x, f(x))$ values in red
- Estimates:
 $(y, f(y))$ values in blue
- Estimated continuous-domain function is piecewise constant

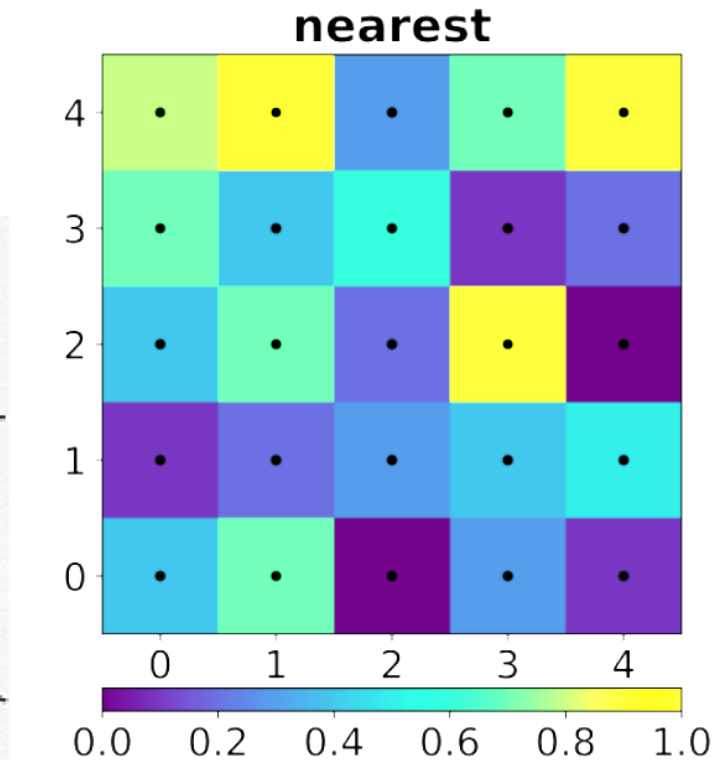


Digital Image Interpolation: Nearest-Neighbor

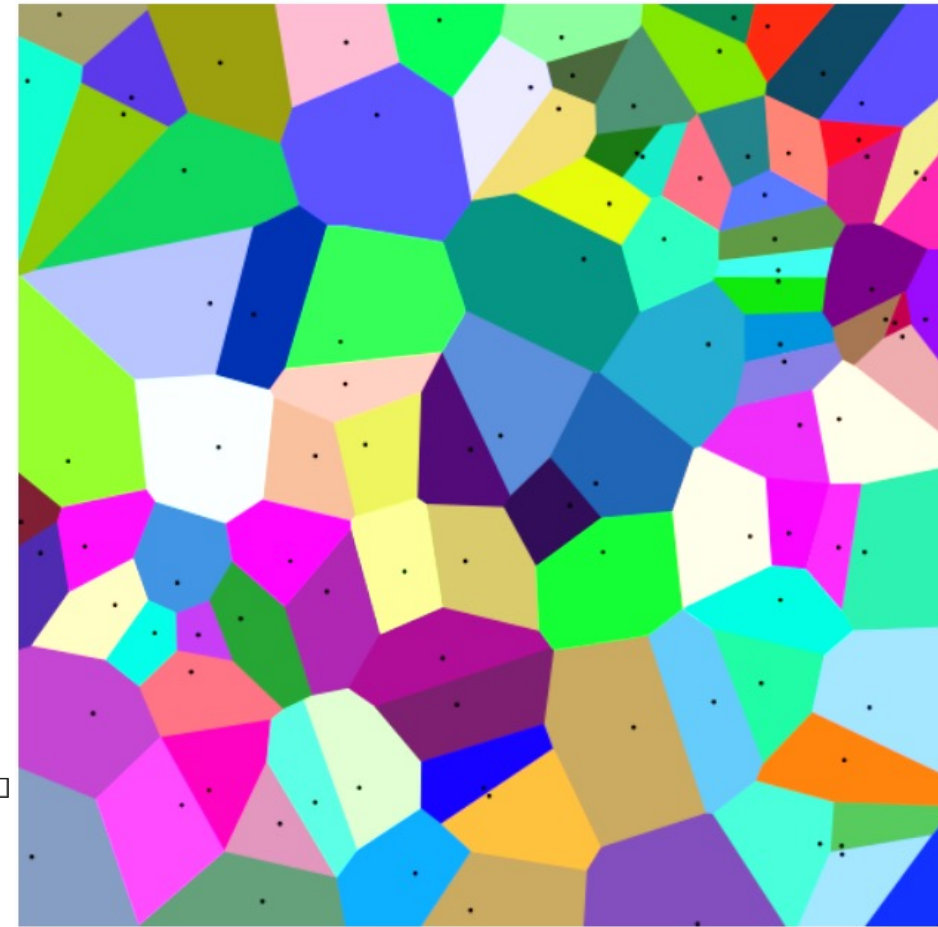
- Nearest-neighbor interpolation in 2D



- Also works on unstructured grid



Nearest neighbor interpolation on a uniform 2D grid (black points). Each colored cell indicates the area in which all the points have the black point in the cell as their nearest black point.



This [Voronoi diagram](#) is an example of nearest neighbor interpolation of a random set of points (black dots) in 2D.

Georgy Voronoy

- Russian mathematician of Ukrainian descent
- At age 40, Voronoy began having stomach problems
 - “Yesterday I had for the first time a clear idea of the algorithm in the theory of forms I am investigating, but also suffered a strong attack of bilious colic which prevented me from working in the evening and from sleeping the whole night. I am so afraid that the results of my enduring efforts, obtained with such difficulty, will perish along with me.”
- Died that year
- Had great influence on Boris Delaunay → Delaunay triangulation
 - Dual of Voronoi diagram
- Son Yurii Voronyi, Ukrainian surgeon, performed world's first human-to-human kidney transplant in 1933

Digital Image Interpolation

- Interpolation of small images



- en.wikipedia.org/wiki/Comparison_gallery_of_image_scaling_algorithms#

- Original large/high-res.



- Nearest neighbor.

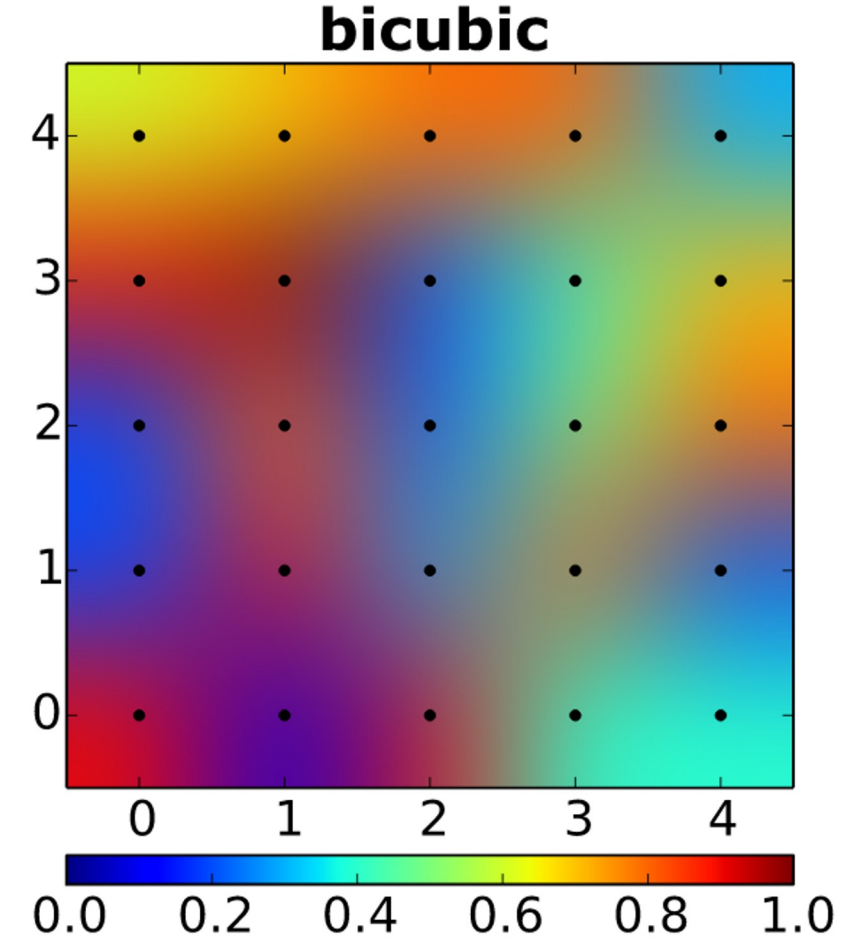
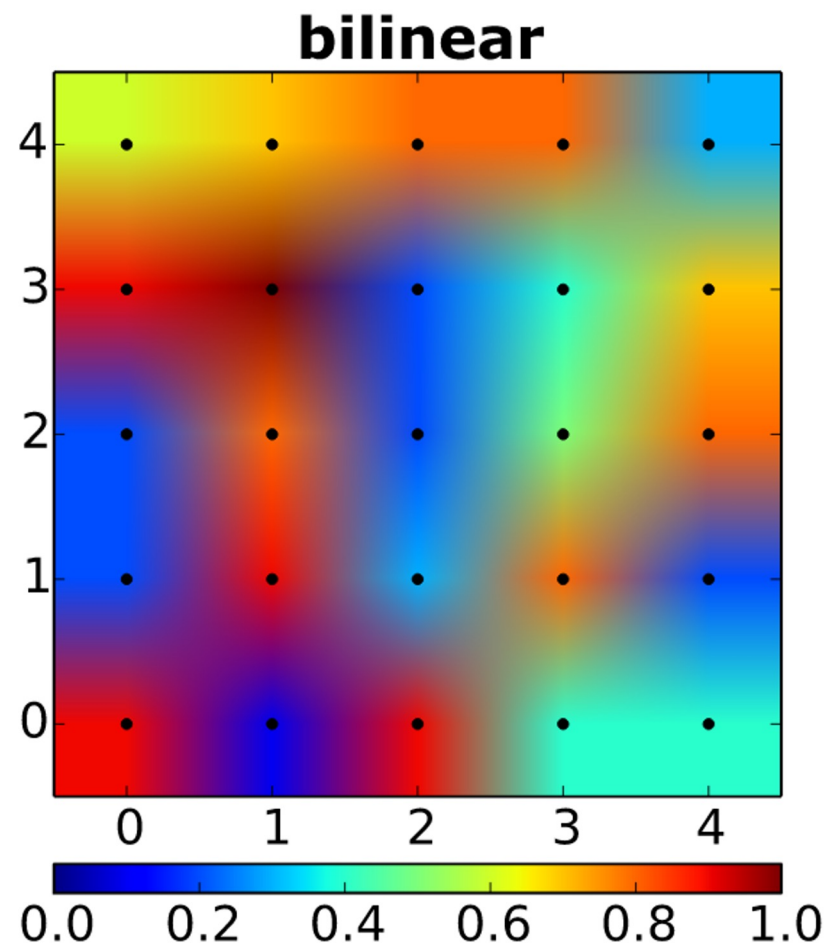
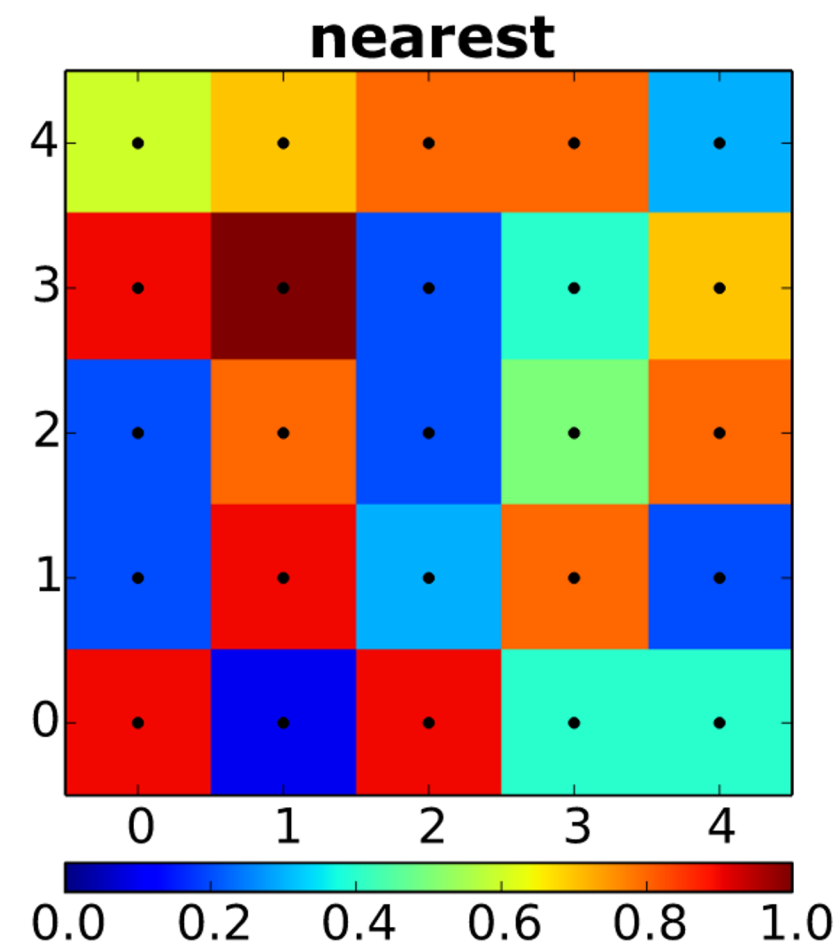


- Bilinear.



Digital Image Interpolation

- Interpolation
 - Bilinear interpolation gives smoother function estimate than nearest neighbor
 - Can we use higher-order polynomials within each piece ? Yes, e.g., bicubic



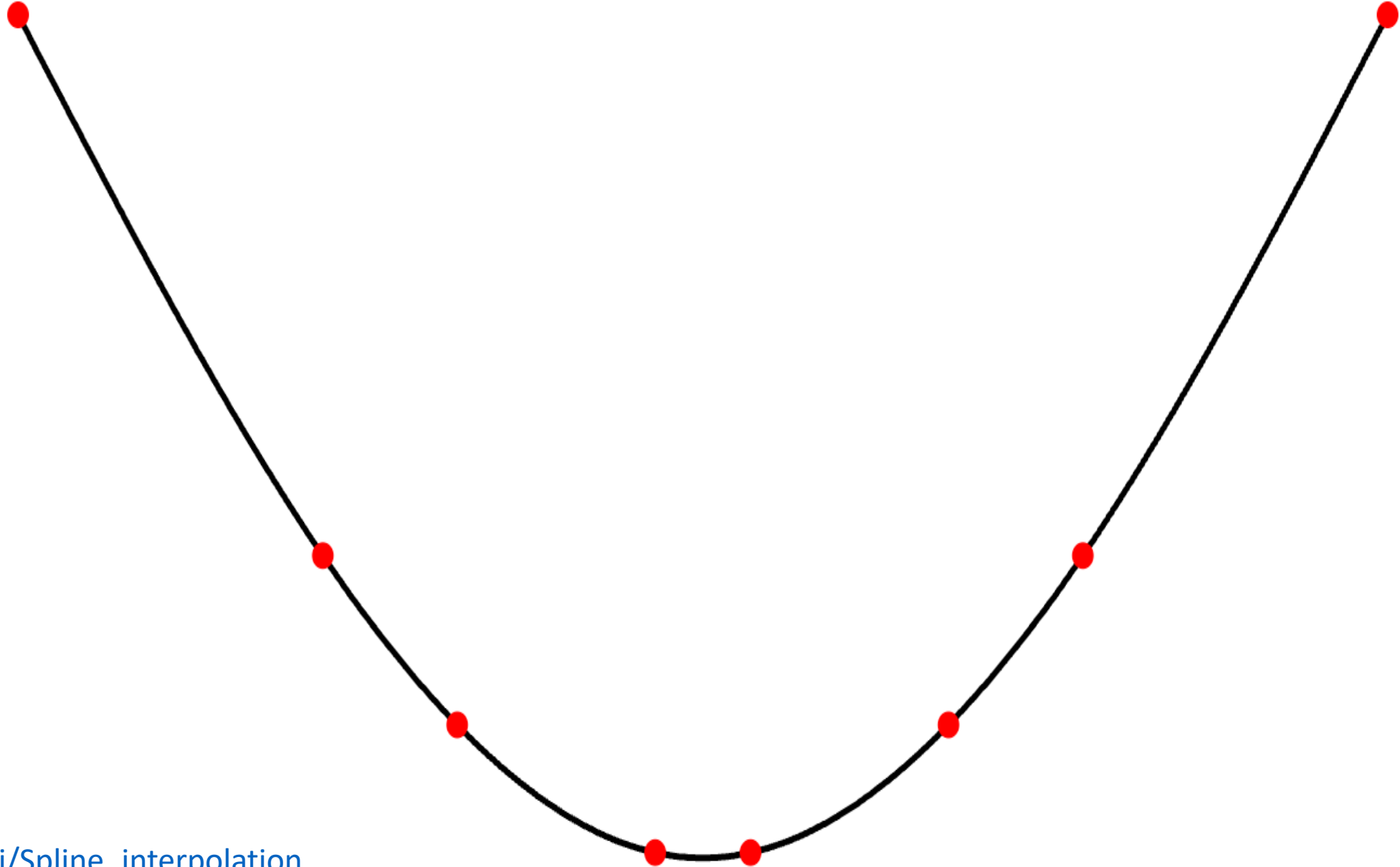
Digital Image Interpolation: Spline Interpolation

- **Spline** (English) =
“a long, thin piece of wood or metal used in building”
 - dictionary.cambridge.org/dictionary/english/spline
- Term “spline” comes from flexible spline devices used by shipbuilders & draftsmen to draw smooth shapes passing via predefined points/“knots”



Digital Image Interpolation: Spline Interpolation

- In our context:
spline =
function
defined
piecewise
by
polynomials



Digital Image Interpolation: Quadratic Spline

- 1D case. **N intervals. Quadratic** polynomial on **each interval**.
- Each quadratic ($a_i + b_i x + c_i x^2$) has 3 parameters $\rightarrow$ total $3N$ unknowns
- **Constraints: within each interval**
 - Value of quadratic must match data at left end $\rightarrow N$ constraints
 - Value of quadratic must match data at right end $\rightarrow N$ constraints
- **Constraints: at joints of intervals (total $N-1$ joints)**
 - First derivative of left & right quadratic must match $\rightarrow N-1$ constraints
- **Constraints: boundary conditions**
 - e.g., assume linearity for either first spline or last spline $\rightarrow 1$ constraint

Digital Image Interpolation: Quadratic Spline

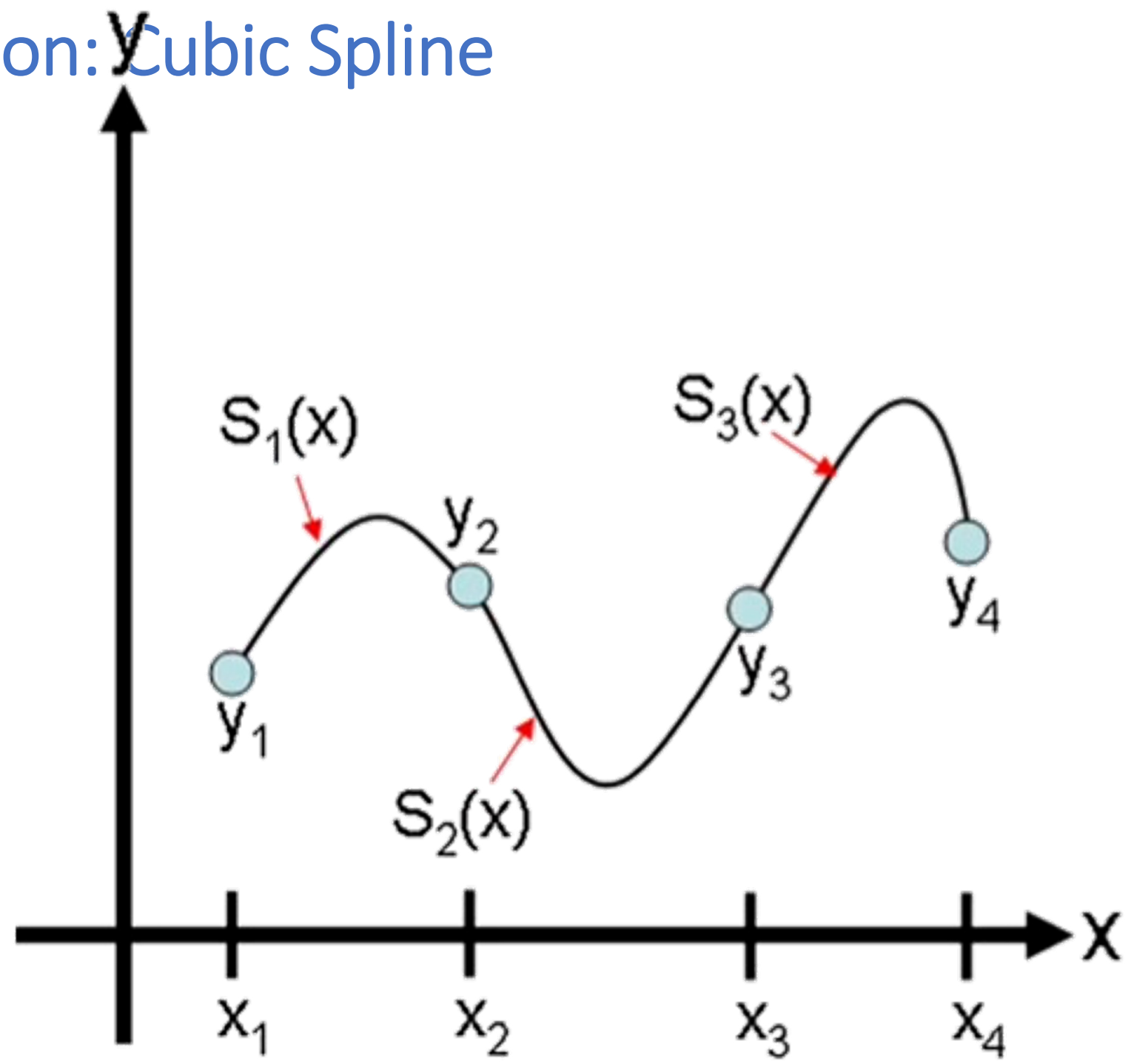
- Data: $(x_1, y_1), (x_2, y_2), \dots, (x_{N+1}, y_{N+1})$

$$\begin{array}{ll} a_1 + b_1 x_1 + c_1 x_1^2 = y_1 & b_1 + 2c_1 x_2 - b_2 - 2c_2 x_2 = 0 \\ a_1 + b_1 x_2 + c_1 x_2^2 = y_2 & b_2 + 2c_2 x_3 - b_3 - 2c_3 x_3 = 0 \\ a_2 + b_2 x_2 + c_2 x_2^2 = y_2 & \vdots \\ a_2 + b_2 x_3 + c_2 x_3^2 = y_3 & \\ \vdots & \end{array}$$

$2c_1 = 0$

Digital Image Interpolation: Cubic Spline

- Cubic spline interpolation

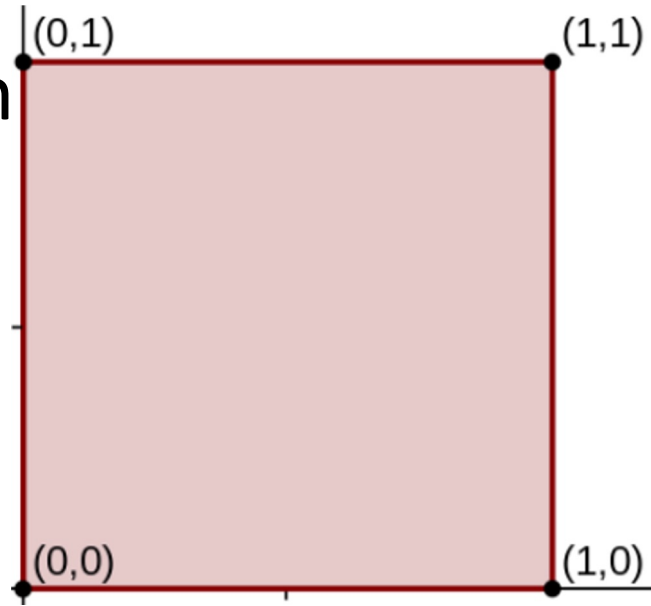


Digital Image Interpolation: Cubic Spline

- 1D case. **N intervals. Cubic** on each **interval**.
- Each cubic ($a_i + b_i x + c_i x^2 + d_i x^3$) has 4 parameters $\rightarrow$ total $4N$ unknowns
- **Constraints: within each interval**
 - Value of cubic must match data at left end & right end $\rightarrow N+N$ constraints
- **Constraints: at joints of intervals (total $N-1$ joints)**
 - First derivative of left and right cubic must match $\rightarrow N-1$ constraints
 - Second derivative of left and right cubic must match $\rightarrow N-1$ constraints
- **Constraints: boundary conditions**
 - e.g., design first and last polynomial to be quadratic ($d_1 = 0, d_N = 0$)
 - e.g., “natural” spline:
at left end of first interval specify 2nd derivative = 0 (curve $\rightarrow$ linear at end),
at right end of last interval specify 2nd derivative = 0 (curve $\rightarrow$ linear at end)

Digital Image Interpolation: Bicubic

- Given: Unit square with corner locations (0,0), (0,1), (1,0), (1,1).
Values of function $f(x,y)$ & derivatives $f_x(x,y)$, $f_y(x,y)$, $f_{xy}(x,y)$ at 4 corners.
- Interpolated surface is a bicubic
 - 16 parameters
- At each **corner**, match **function values** (4 equations)
- At each **corner**, match function **derivatives** (3x4 equations)
- 16 linear equations. Solve.



$$p(x, y) = \sum_{i=0}^3 \sum_{j=0}^3 a_{ij} x^i y^j$$

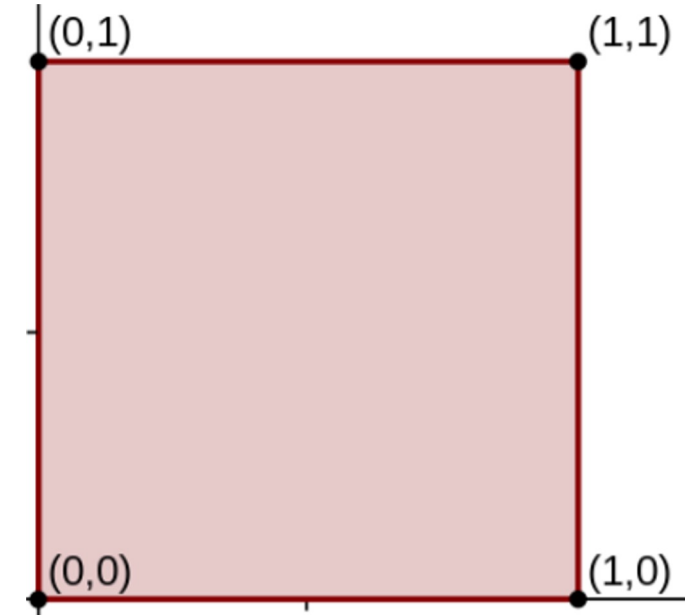
$$p_x(x, y) = \sum_{i=1}^3 \sum_{j=0}^3 a_{ij} i x^{i-1} y^j,$$

$$p_y(x, y) = \sum_{i=0}^3 \sum_{j=1}^3 a_{ij} x^i j y^{j-1},$$

$$p_{xy}(x, y) = \sum_{i=1}^3 \sum_{j=1}^3 a_{ij} i x^{i-1} j y^{j-1}$$

Digital Image Interpolation: Bicubic

- How to get function values and function-derivative values at corners ?
- Standard numerical schemes using finite-difference approximations
- e.g., $f_x(x,y) := 0.5 (f(x+1,y) - f(x-1,y))$
- e.g., $f_y(x,y) := 0.5 (f(x,y+1) - f(x,y-1))$
- e.g., $f_{xy}(x,y)$ can be obtained by finite-difference approximation of y derivative on $f_x(x,y)$, i.e.,
 $f_{xy}(x,y) := 0.25$
 $([f(x+1,y+1) + f(x-1,y-1)] - [f(x-1,y+1) + f(x+1,y-1)]) \dots$
- ... OR
finite-difference approximation of x derivative on $f_y(x,y)$



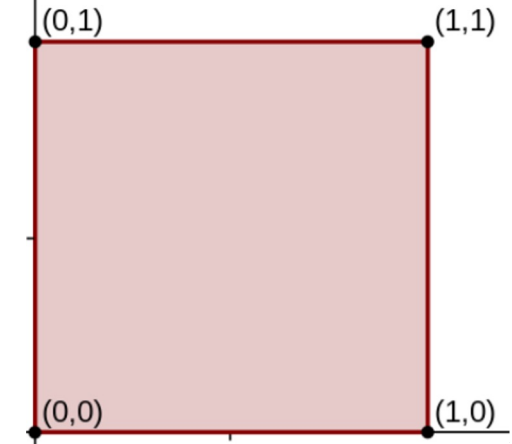
Digital Image Interpolation: Bicubic

- What about continuity of interpolated function across shared boundary ?
- Consider unit square. Consider interpolated function $p(x,y)$ over line segment joining (0,0) to (1,0)

1) $p(x, y=0)$,
 $p_x(x, y=0)$,
 $p_y(x, y=0)$,
 $p_{xy}(x, y=0)$
depend on which coefficients ?

- Coefficient used only when
 $j=0$ in $p(\cdot)$ & $p_x(\cdot)$ $j=1$ in $p_y(\cdot)$ & $p_{xy}(\cdot)$

• $\{a_{00}, a_{10}, a_{20}, a_{30}\} \cup \{a_{01}, a_{11}, a_{21}, a_{31}\}$



$$p(x, y) = \sum_{i=0}^3 \sum_{j=0}^3 a_{ij} x^i y^j$$

$$p_x(x, y) = \sum_{i=1}^3 \sum_{j=0}^3 a_{ij} i x^{i-1} y^j,$$

$$p_y(x, y) = \sum_{i=0}^3 \sum_{j=1}^3 a_{ij} x^i j y^{j-1},$$

$$p_{xy}(x, y) = \sum_{i=1}^3 \sum_{j=1}^3 a_{ij} i x^{i-1} j y^{j-1}$$

Digital Image Interpolation: Bicubic

- Consider interpolated function $p(x,y)$ over line segment joining $(0,0)$ to $(1,0)$
- 2) How do $\{a_{00}, a_{10}, a_{20}, a_{30}\} \cup \{a_{01}, a_{11}, a_{21}, a_{31}\}$ depend on data involving $(0,0)$ & $(1,0)$?
- Consider equations satisfied during bicubic fit involving $\{a_{00}, a_{10}, a_{20}, a_{30}\} \searrow$
- Consider equations satisfied during bicubic fit involving $\{a_{01}, a_{11}, a_{21}, a_{31}\} \swarrow$
- Those 8 coefficients can be inferred (uniquely) from data at $(0,0)$ & $(1,0)$

$$p(x, y) = \sum_{i=0}^3 \sum_{j=0}^3 a_{ij} x^i y^j$$

$$p_x(x, y) = \sum_{i=1}^3 \sum_{j=0}^3 a_{ij} i x^{i-1} y^j,$$

$$p_y(x, y) = \sum_{i=0}^3 \sum_{j=1}^3 a_{ij} x^i j y^{j-1},$$

$$p_{xy}(x, y) = \sum_{i=1}^3 \sum_{j=1}^3 a_{ij} i x^{i-1} j y^{j-1}$$

$$f(0, 0) = p(0, 0) = a_{00},$$

$$f(1, 0) = p(1, 0) = a_{00} + a_{10} + a_{20} + a_{30}$$

$$f_x(0, 0) = p_x(0, 0) = a_{10},$$

$$f_x(1, 0) = p_x(1, 0) = a_{10} + 2a_{20} + 3a_{30}$$

$$f_y(0, 0) = p_y(0, 0) = a_{01},$$

$$f_y(1, 0) = p_y(1, 0) = a_{01} + a_{11} + a_{21} + a_{31}$$

$$f_{xy}(0, 0) = p_{xy}(0, 0) = a_{11},$$

$$f_{xy}(1, 0) = p_{xy}(1, 0) = a_{11} + 2a_{21} + 3a_{31}$$

Digital Image Interpolation: Bicubic

- Bicubic interpolation
- So far, we analyzed interpolated function over line segment joining $(0,0)$ to $(1,0) \rightarrow$ along X axis to find that $p(x,0)$, $p_x(x,0)$, $p_y(x,0)$, $p_{xy}(x,0)$ depend only on data at $(0,0)$ & $(1,0)$
- This will also hold true for other unit square with corners at $(0,0)$, $(1,0)$, $(1,-1)$, $(0,-1)$
- So, $p(x,y)$ and its partial derivatives with respect to x , y , xy are continuous across line segment
- But not necessarily second (non-mixed) partial derivatives, i.e., p_{xx} , p_{yy}
- Similar continuity properties will exist along all edges of square

Digital Image Interpolation: Tricubic

- Interpolating function
- 64 coefficients

$$f(x, y, z) = \sum_{i=0}^3 \sum_{j=0}^3 \sum_{k=0}^3 a_{ijk} x^i y^j z^k$$

- 1) Match function values at cube corners $\rightarrow$ 8 equations
- 2) Match function 1st derivatives (w.r.t. x, y, z) at cube corners
 $\rightarrow 8*3=24$ equations
- 3) Match function 2nd mixed derivatives (w.r.t. xy, yz, zx) at cube corners
 $\rightarrow 8*3=24$ equations
- 4) Match function 3rd mixed derivative (w.r.t. xyz) at cube corners
 $\rightarrow 8$ equations

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Tricubic interpolation in three dimensions

F. Lekien^{*,†,1,2} and J. Marsden²

¹Mechanical and Aerospace Engineering, Princeton University, U.S.A.

²Control and Dynamical Systems, California Institute of Technology, U.S.A.

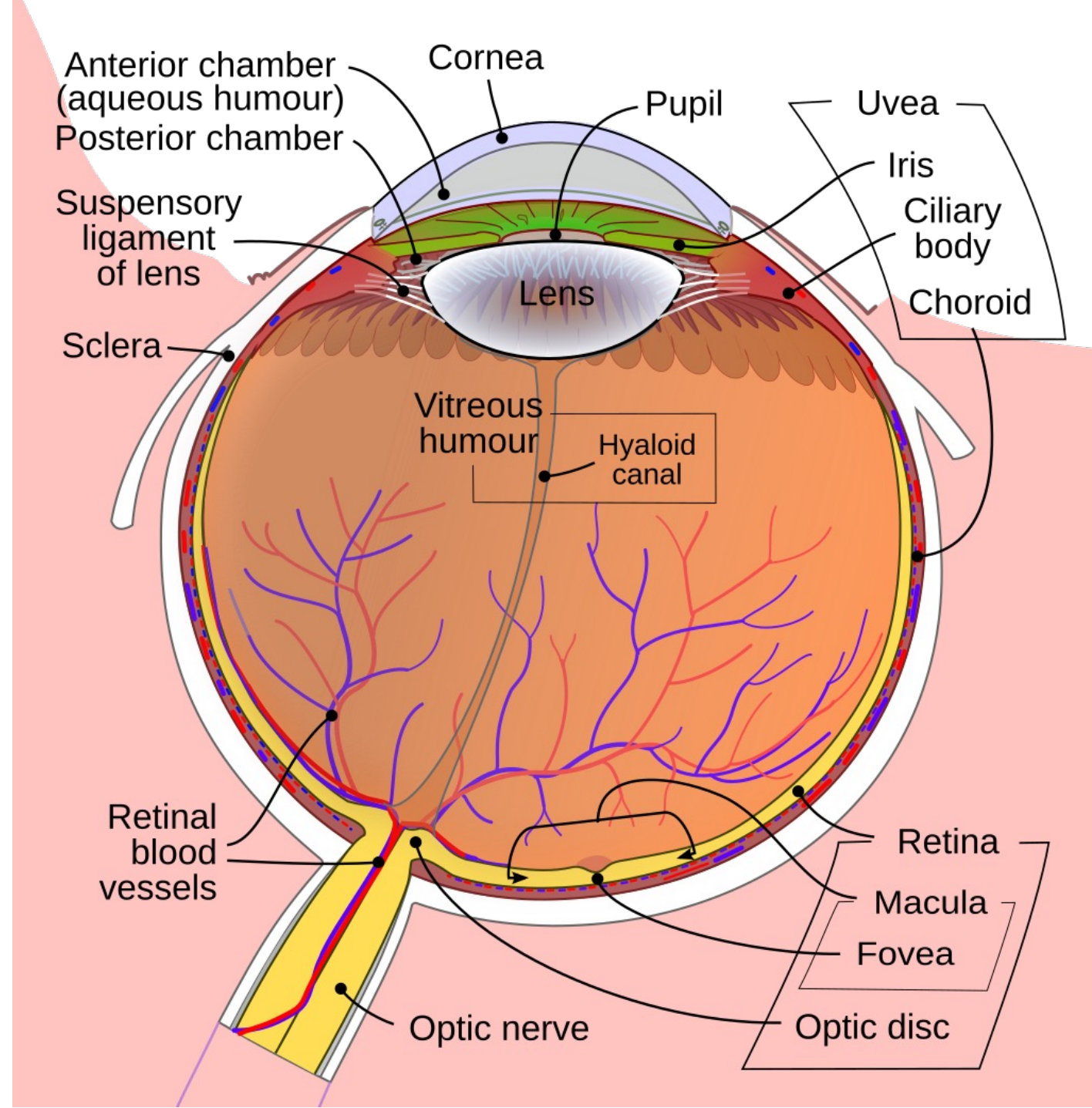
Digital Image Interpolation

- Image resizing
- Example 1: reduce image size
 - $301 \times 301 \rightarrow 101 \times 101$ may not need interpolation
 - $101 \times 101 \rightarrow 31 \times 31$ needs interpolation
 - Reducing image size
may introduce moire artifacts
- Example 2: increase image size along each dimension
 - Needs interpolation

Human Visual System

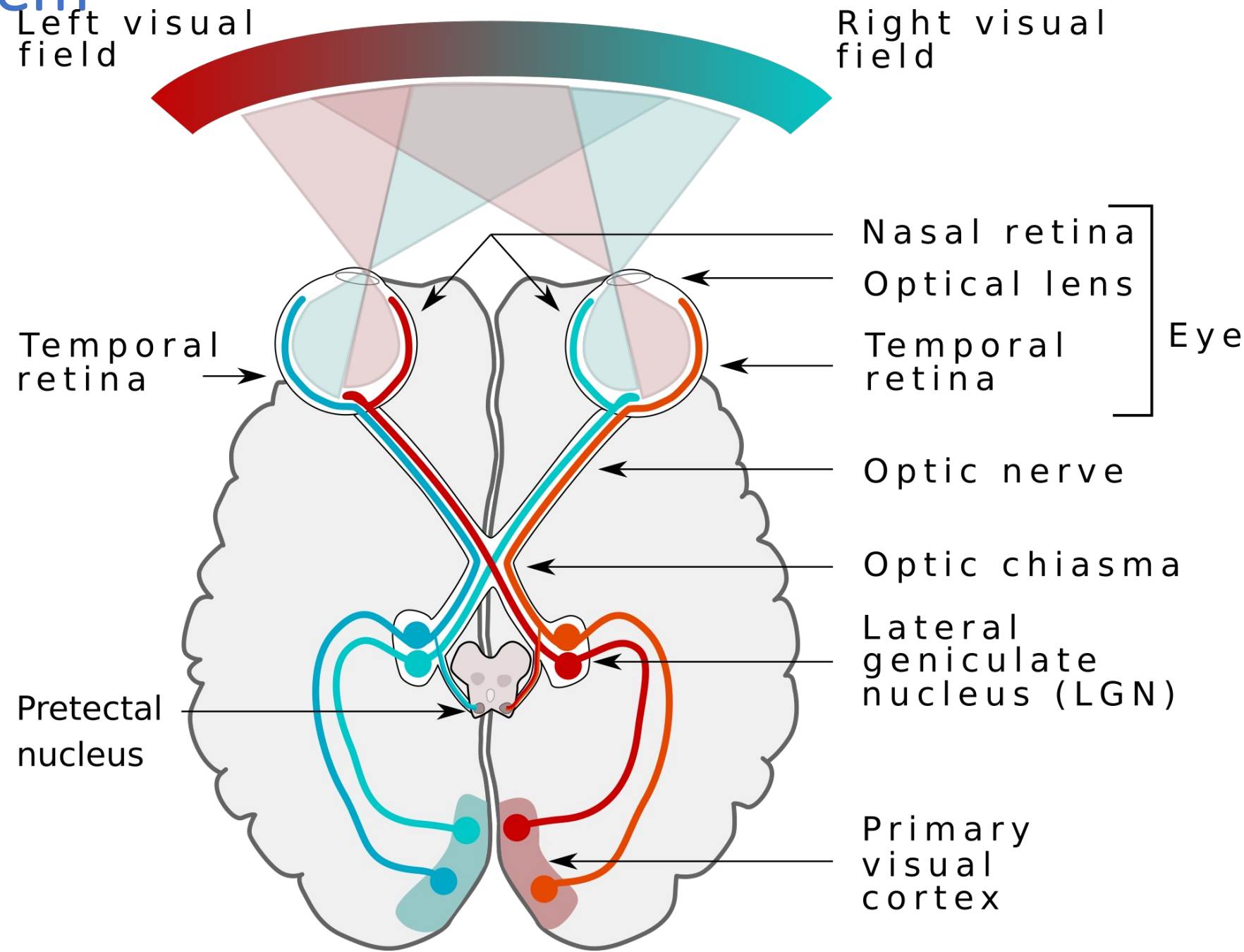
- Human eye

- en.wikipedia.org/wiki/Visual_system



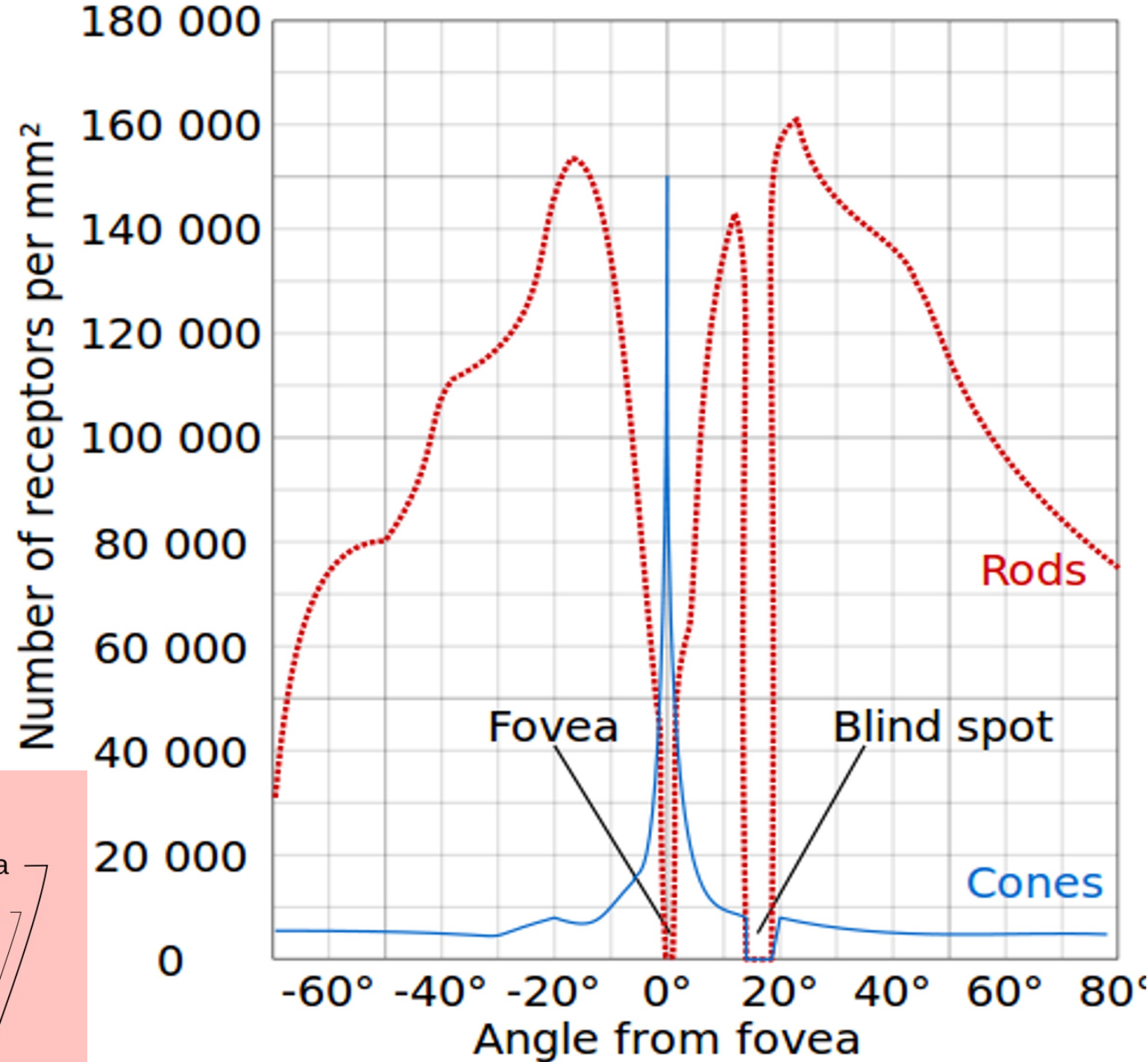
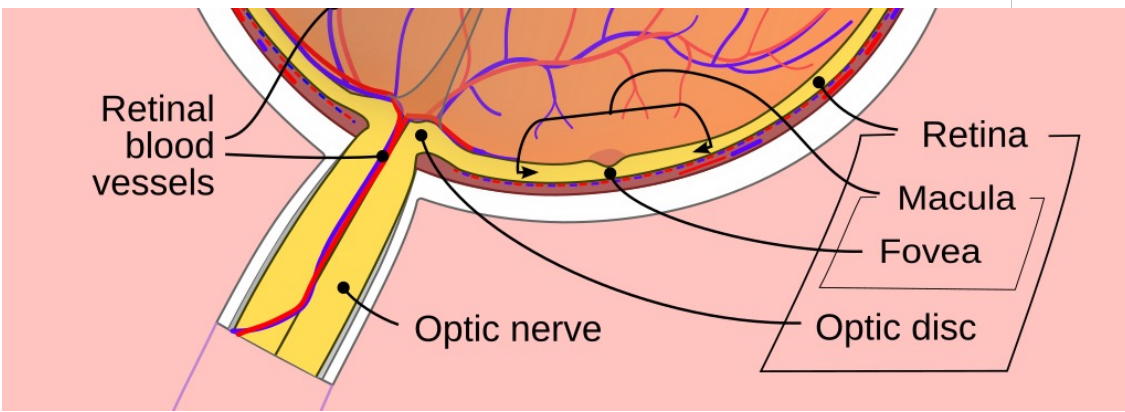
Human Visual System

- Visual interpretation in human brain



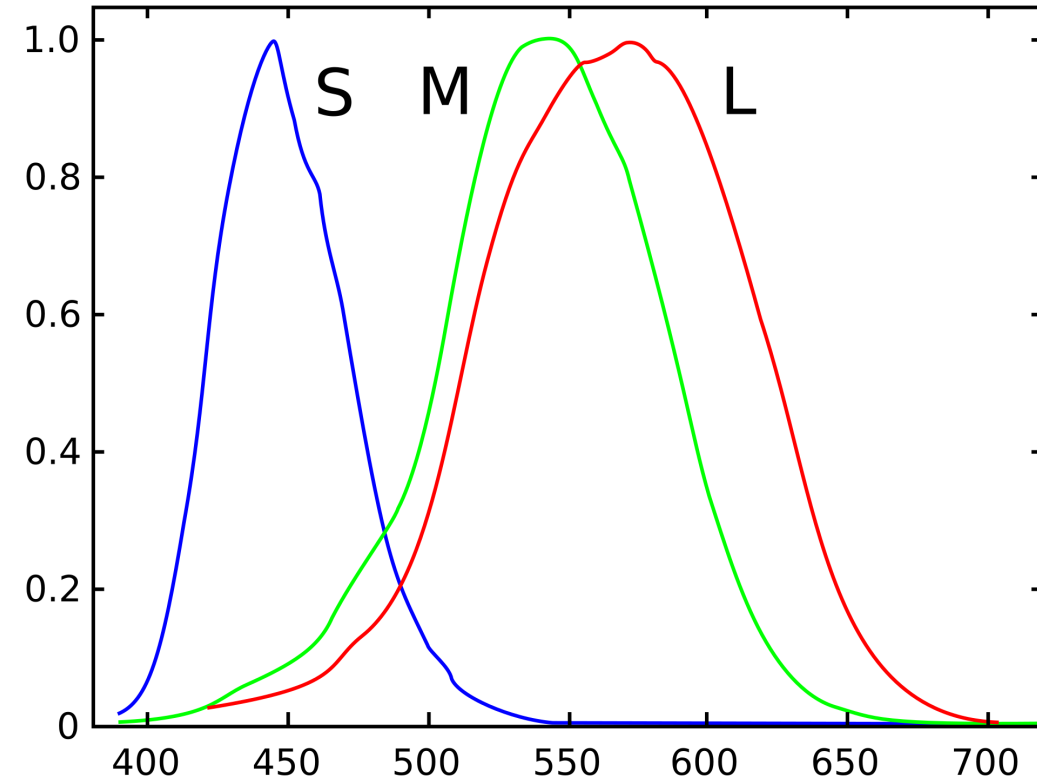
Human Visual System

- Human retinal cells
 - Cones → Color vision, visual acuity
 - Rods → Low-light vision, more sensitive than cones to intensity but NOT color



Human Visual System

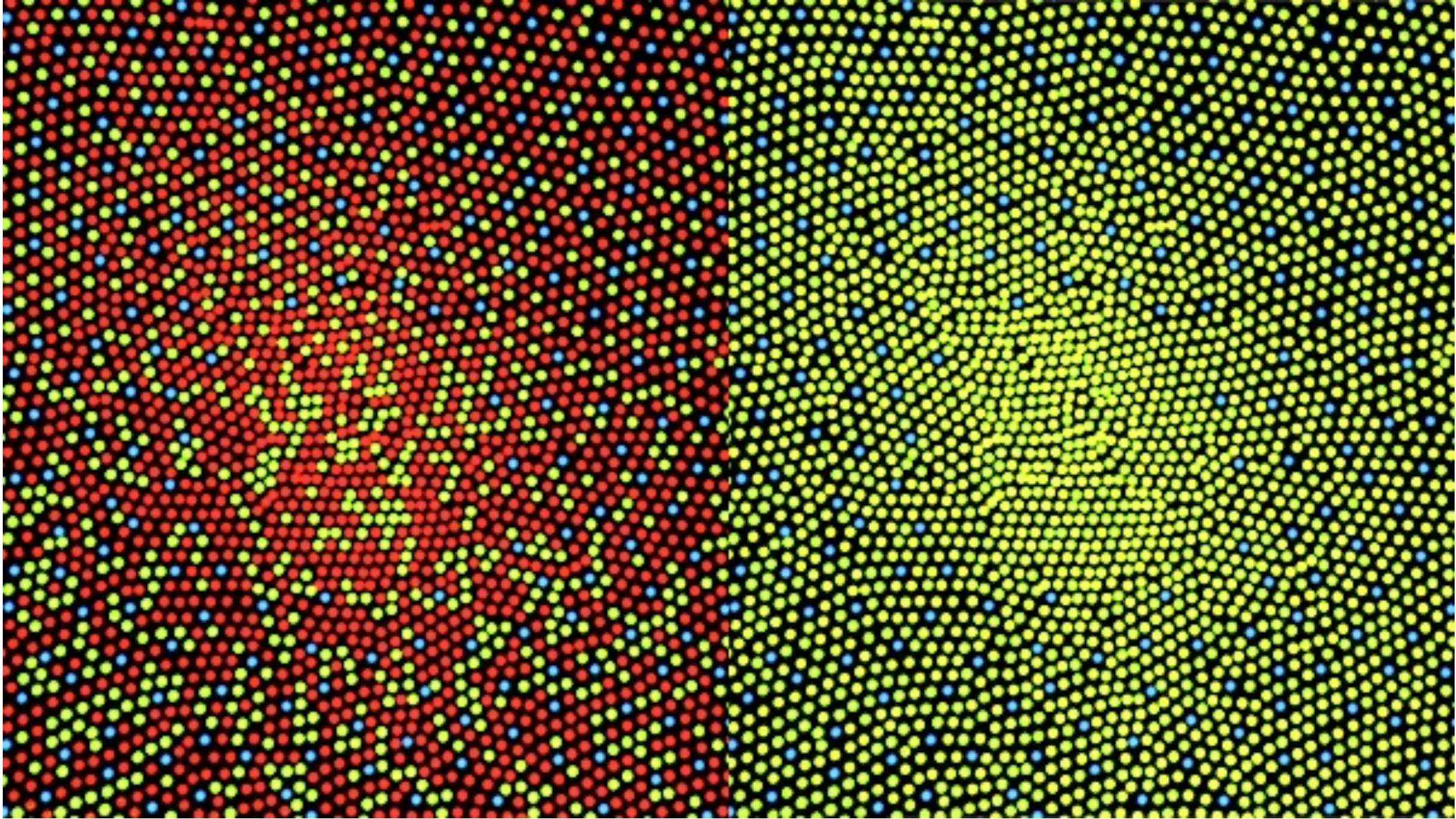
- Human color vision is trichromatic
- Based on color-triangle theory proposed by Thomas Young; extended by James Clerk Maxwell, Hermann von Helmholtz
- 3 different types of cone cells in human eye →
3 independent channels for conveying color information
- 3 types of cones (L, M, S) have pigments that respond best to light of
long wavelength (around 560 nm),
medium wavelength (around 530 nm),
short wavelength (around 420 nm)
 - Normalized responsivity spectra of human cone cells →



Human Visual System

- Human retina: distribution of cone cells in fovea

- Left:
normal
person
- Right:
color-blind
person



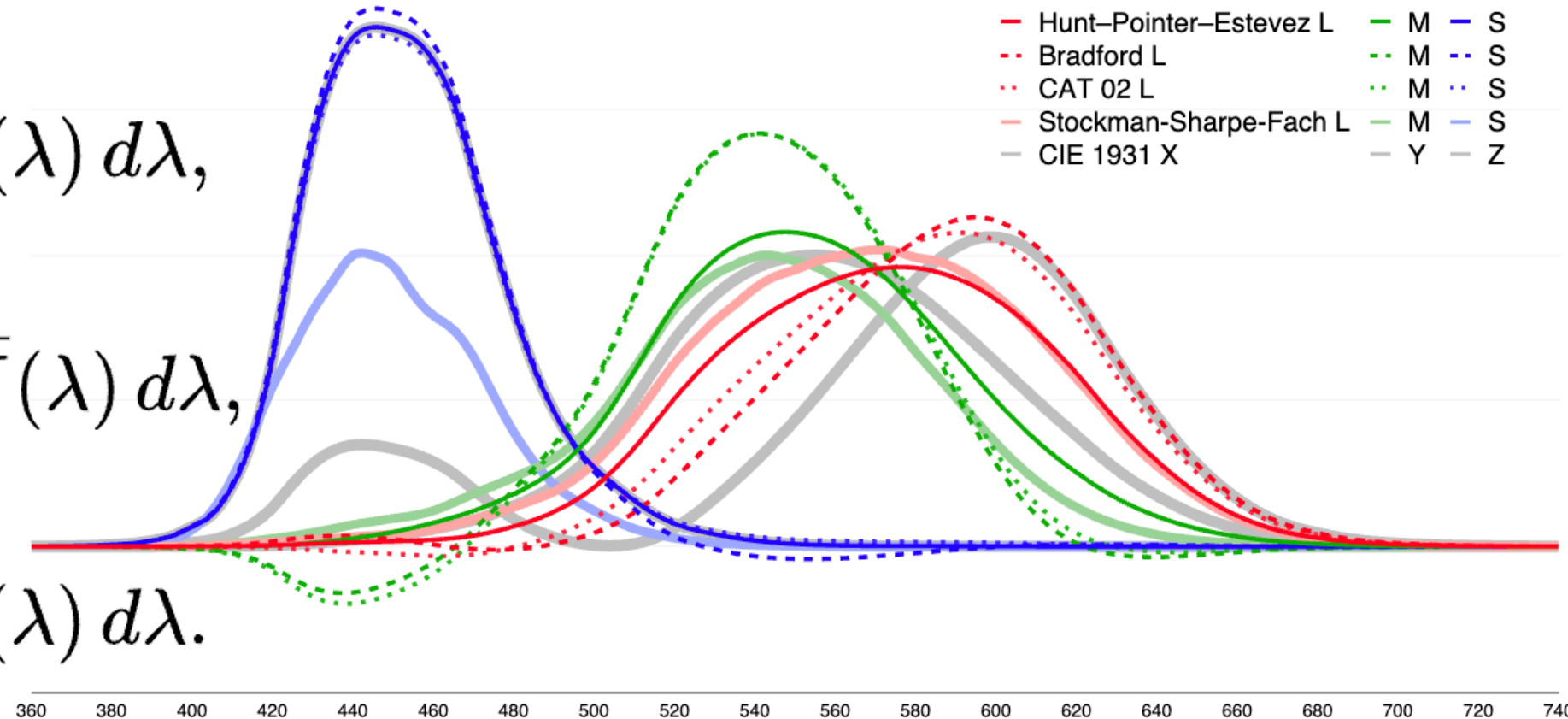
Human Visual System

- “LMS color spaces” (many definitions)
- Given incoming/stimulus light (into eye) of spectral density $S(\lambda)$
- Given spectral sensitivities of L,M,S cones as $\bar{L}(\lambda)$, $\bar{M}(\lambda)$, $\bar{S}(\lambda)$
- L,M,S cone-cell sensors measure:

$$L = \int_0^{\infty} S(\lambda) \bar{L}(\lambda) d\lambda,$$

$$M = \int_0^{\infty} S(\lambda) \bar{M}(\lambda) d\lambda,$$

$$S = \int_0^{\infty} S(\lambda) \bar{S}(\lambda) d\lambda.$$



Human Visual System

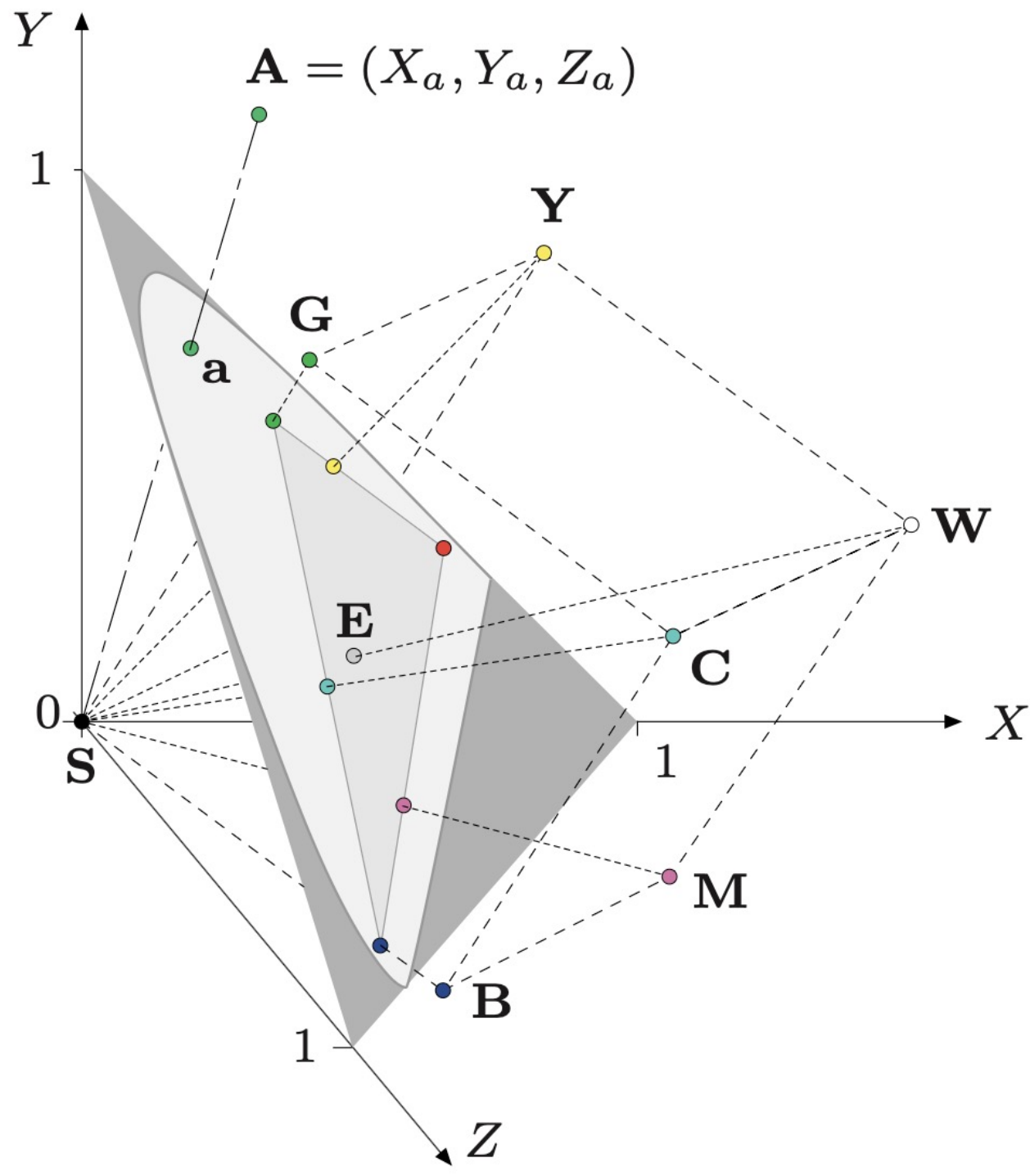


International Commission on Illumination
Commission Internationale de l'Eclairage
Internationale Beleuchtungskommission

- LMS color space relates to CIE XYZ color space
- Designs new spectral sensitivity functions $\bar{x}(\lambda)$, $\bar{y}(\lambda)$, $\bar{z}(\lambda)$...
- Also designs **new primary colors X, Y, Z** such that:
 - **All human-visible colors** (not representable in RGB space) can be modeled as a summation of **positive-only components**
→ all visible colors lie inside a 3D (generalized-)cone shaped region
 - Y component corresponds to perceived lightness or luminosity of a color
- Interestingly, visible colors don't include designed primary colors themselves → designed new primary colors X,Y,Z are **imaginary**
- Linear transformation between LMS and XYZ

Human Visual System

- New X,Y,Z imagined primary colors
- Defines 3 chromaticity values
 - $x := X / (X+Y+Z)$
 - $y := Y / (X+Y+Z)$
 - $z := Z / (X+Y+Z)$
 - White $(x,y,z) := (1/3,1/3,1/3)$
- RGB colors can be mapped to XYZ colors by linear transformation (not vice versa)



Human Visual System

- Visualize x, y . Ignore $z = 1 - x - y$.

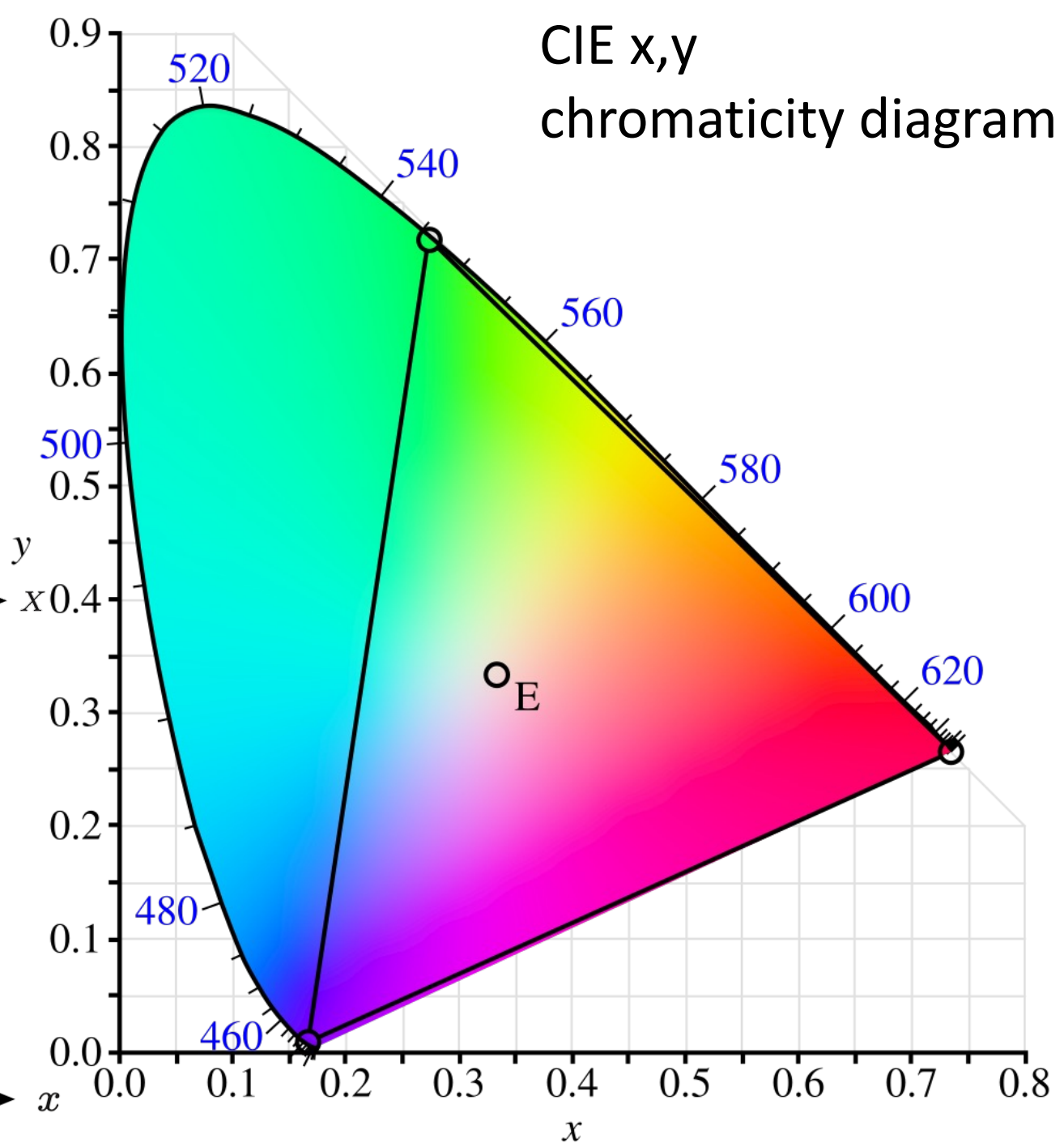
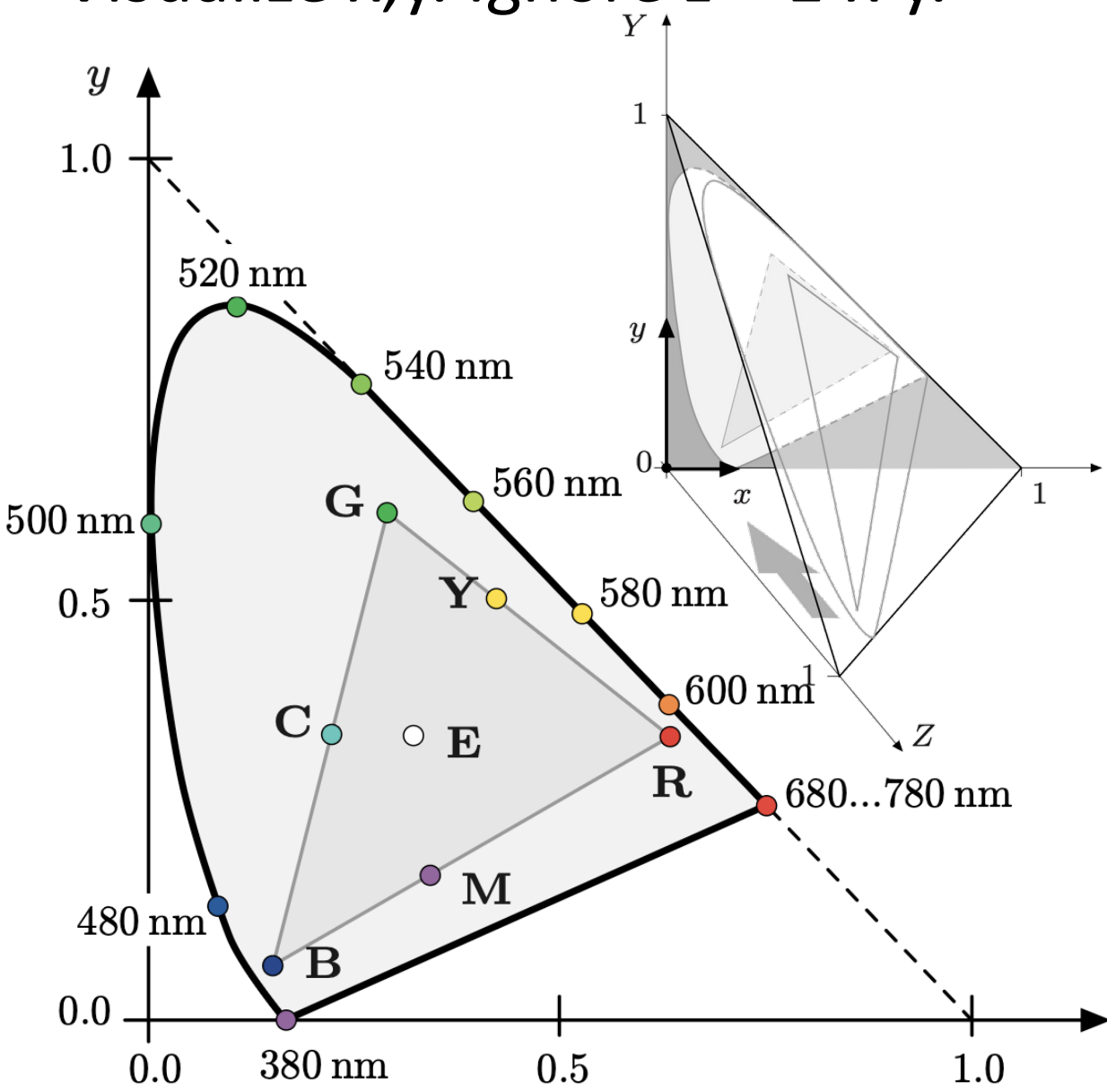
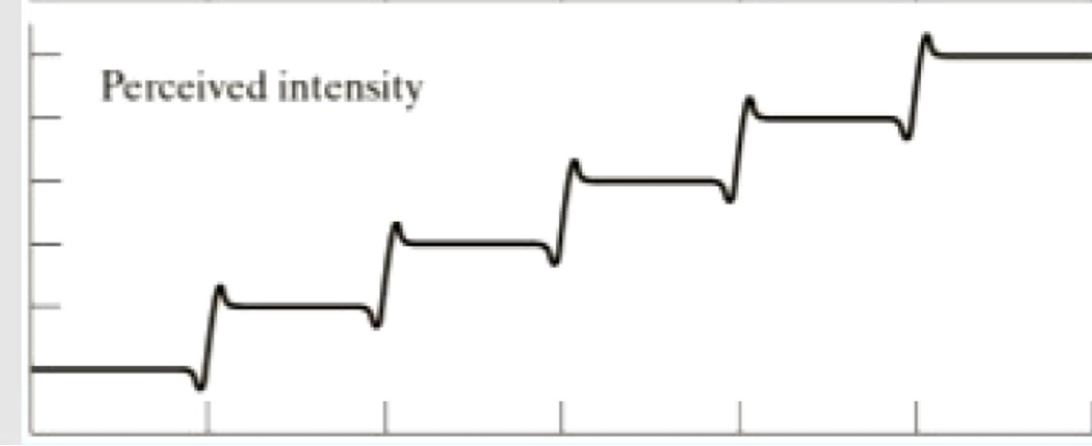
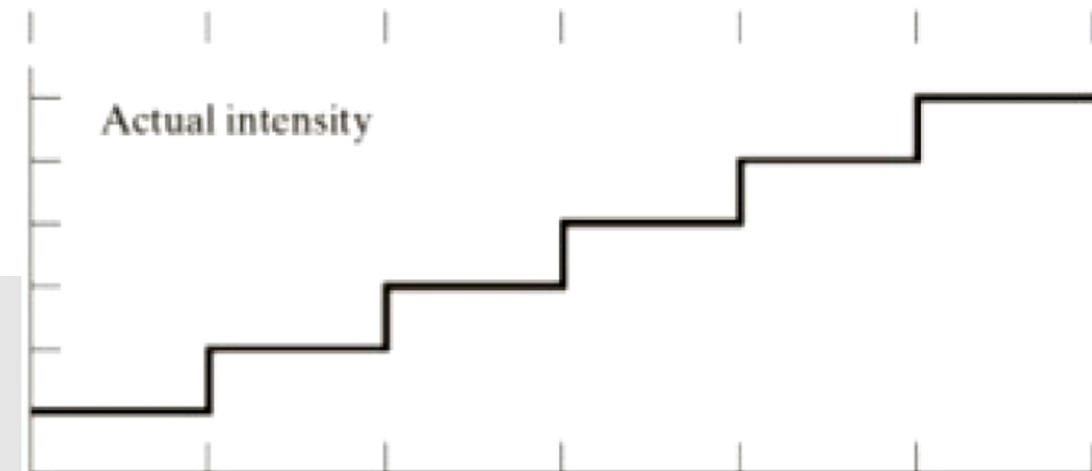


Image Perception: Optical Illusions

- “**Mach band**” effect
 - Perceived intensity NOT a simple function of actual intensity
 - Due to post-processing of visual signals by our visual system
- Exaggerated contrast between edges of slightly differing gray shades appears as soon as they touch



Ernst Mach

- Ernst Waldfried Josef Wenzel Mach
- Austrian physicist
 - Studied shock waves
→ Mach number
- Also worked in physiology, psychology (Mach bands)
- Philosopher:
called
“Buddha of science”

Mach band

Mach diamonds

Mach number

Mach reflection

Mach wave

Mach's principle

Criticism of Newton's
bucket argument^[1]

Empirio-criticism

Oblique effect

Relationalism

Shock waves

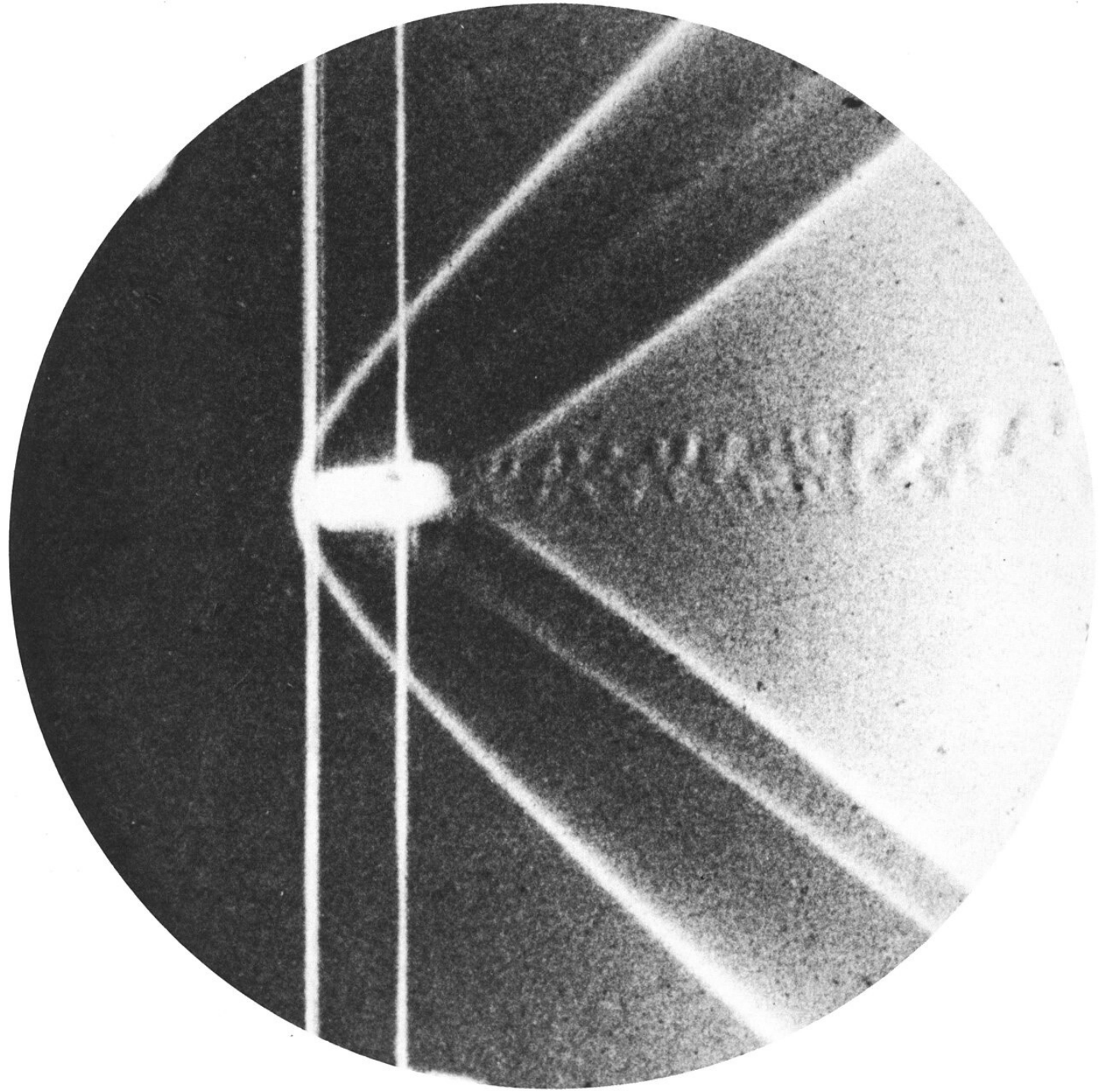
Stereokinetic stimulus

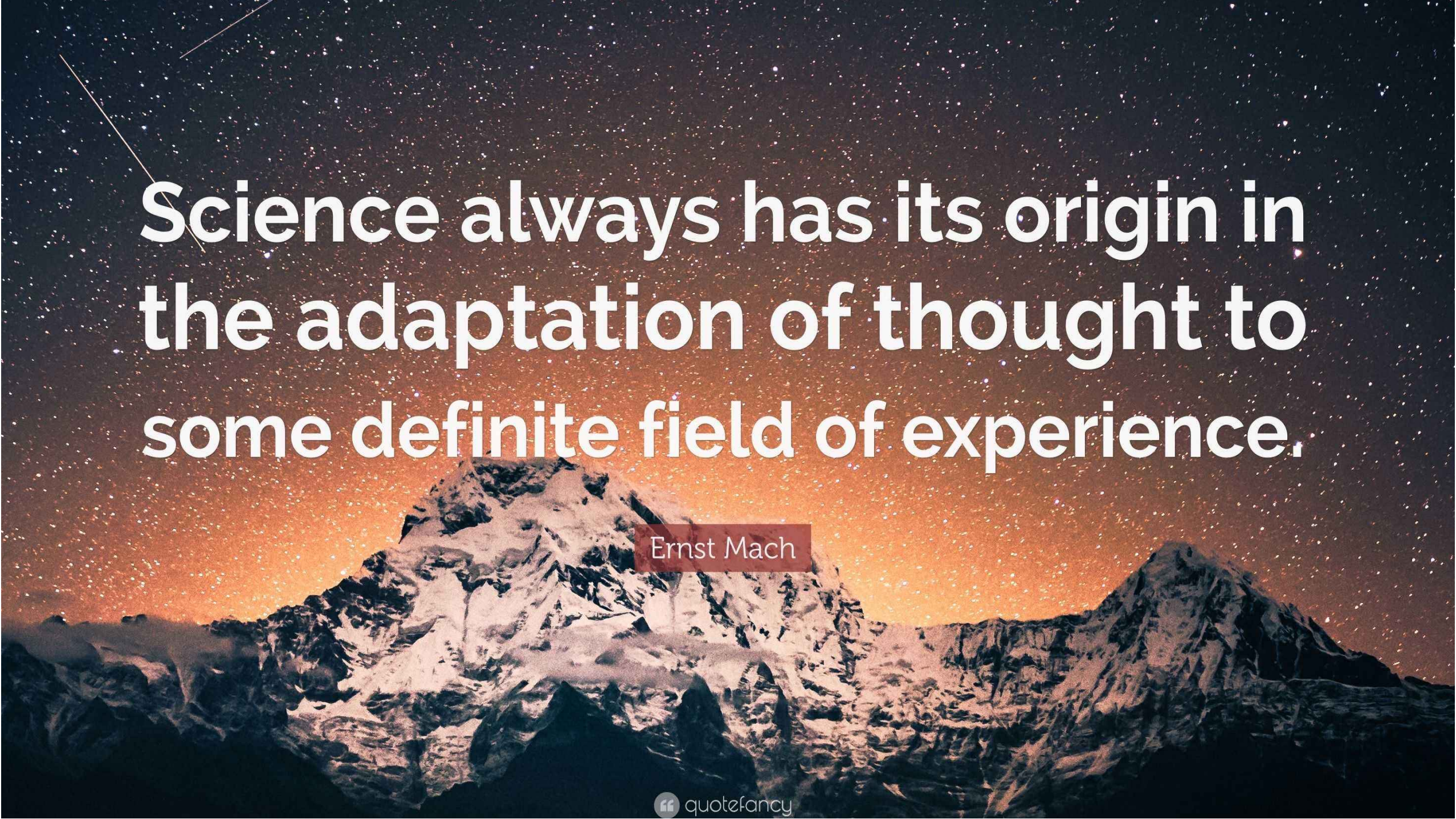
Ludwig Mach



Ernst Mach

- Bullet, shock wave →
- Homeschooled until 14
- Godfather of
Wolfgang Ernst Pauli

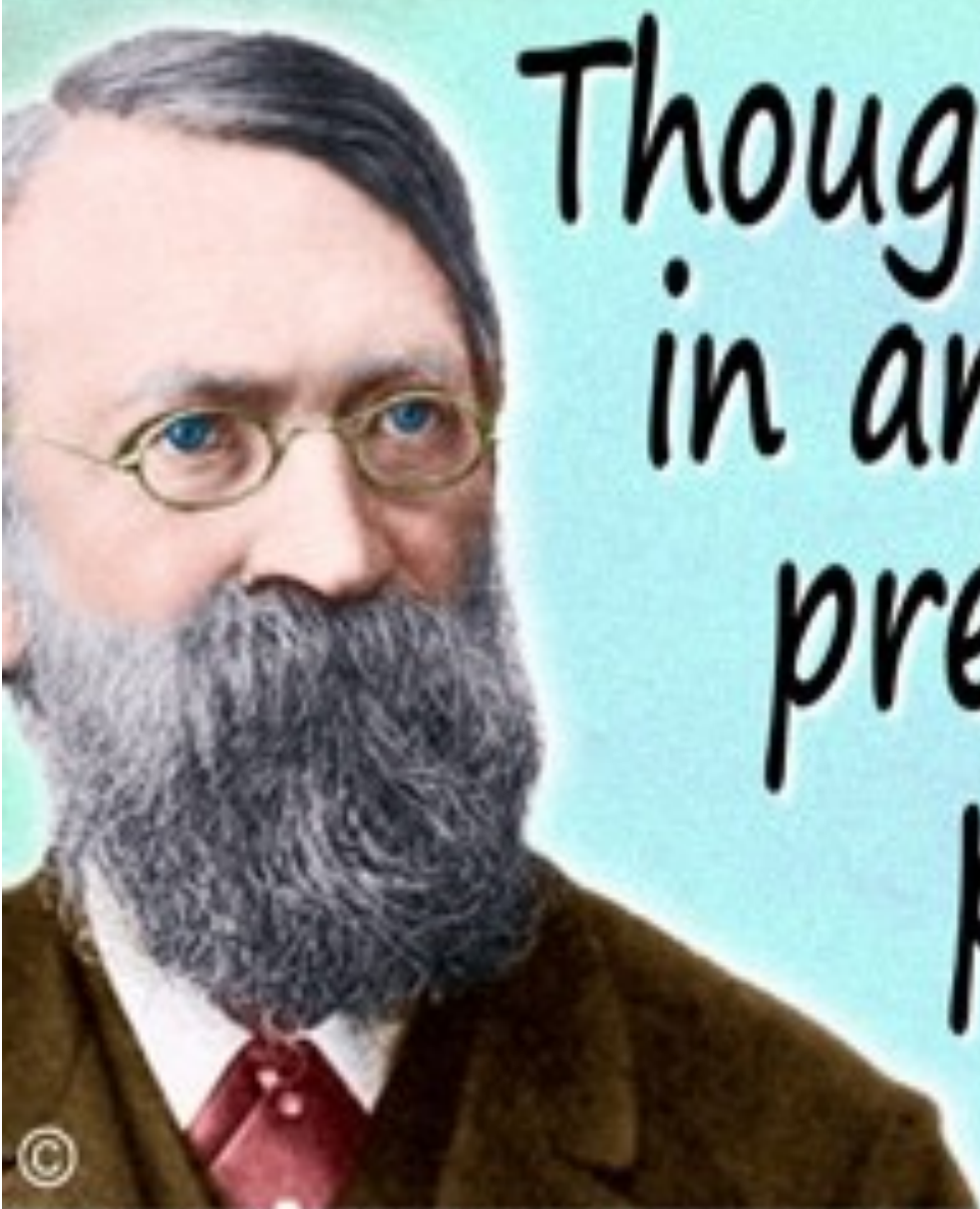




Science always has its origin in
the adaptation of thought to
some definite field of experience.

Ernst Mach

Ernst Mach



Thought experiment is
in any case a necessary
precondition for
physical experiment.

Ernst Mach

Image Perception: Optical Illusions

- Mach band effect in chest X-rays

BMJ Case Reports

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Pseudo-pneumothorax: skin fold is an excellent imitator

Kenichi Kishimoto¹, Takashi Watari¹,  Yasuharu Tokuda²

Image Perception: Optical Illusions

- (A) Chest Xray shows curvilinear line at right lateral zone of right lung
- (B) Chest CT revealed absence of pneumothorax

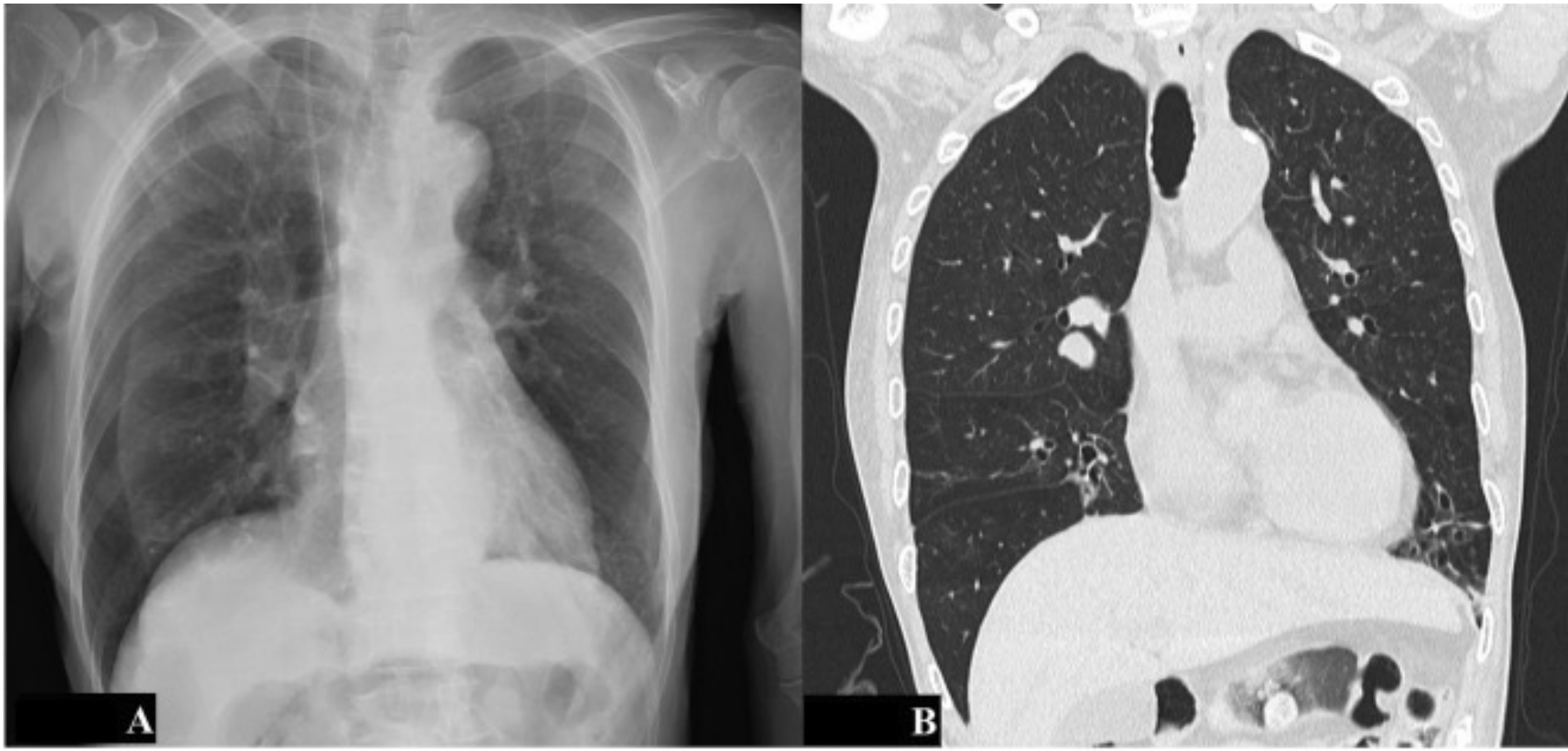


Image Perception: Optical Illusions

- “Dental students are more susceptible than dentists to misinterpreting as a horizontal root fracture (a Mach band illusion) what is known to be the junction of alveolar crestal bone and root.”
- “when presented with a scenario of trauma, both students and dentists are more likely to mistake what is being seen as a fracture line.”

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Effect of Scenario and Experience on Interpretation of Mach Bands

Christen J. Nielsen, DMD, MS

Image Perception: Optical Illus

- Simultaneous contrast

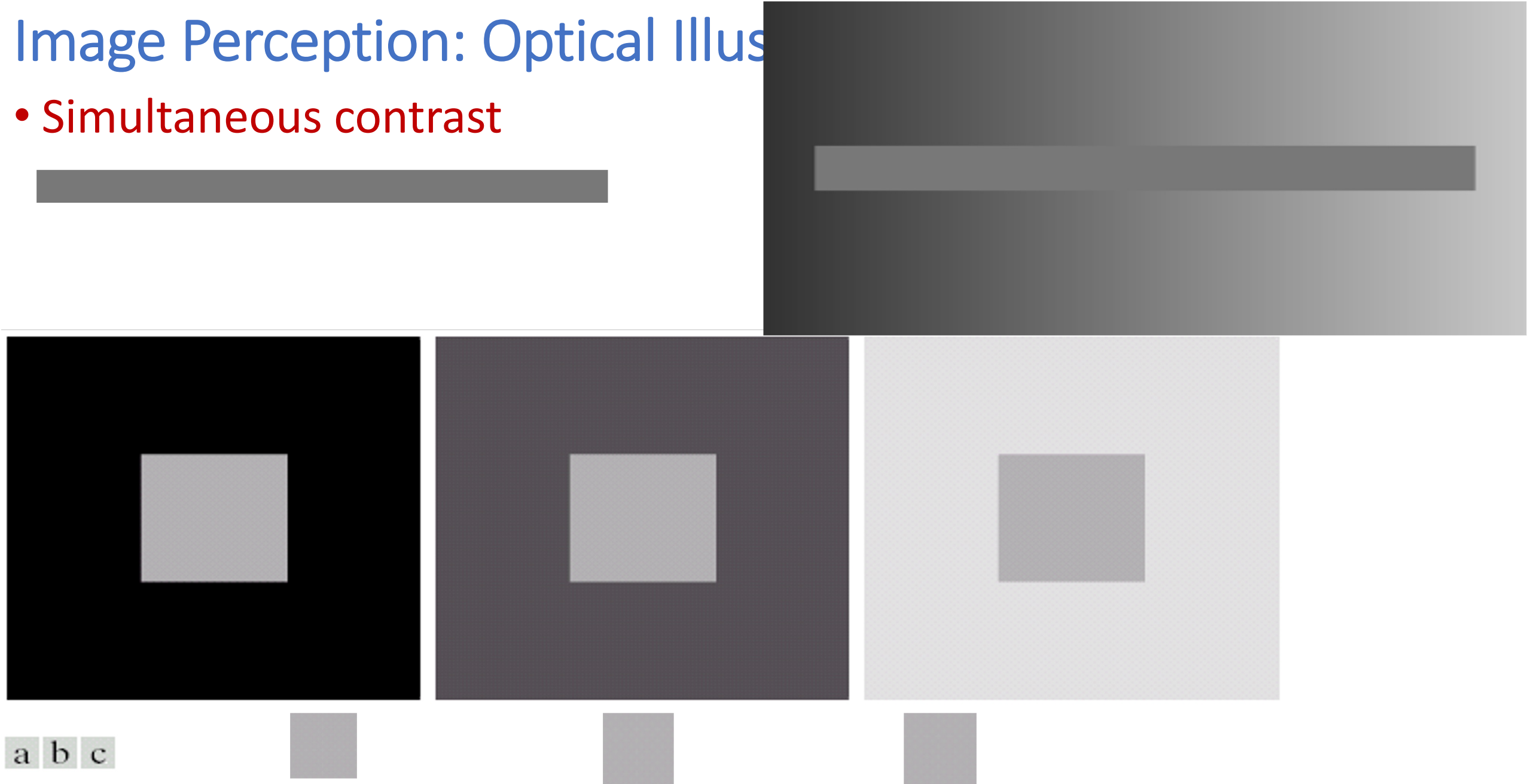


FIGURE 2.8 Examples of simultaneous contrast. All the inner squares have the same intensity, but they appear progressively darker as the background becomes lighter.

Image Perception: Optical Illusions

- Filling in missing data

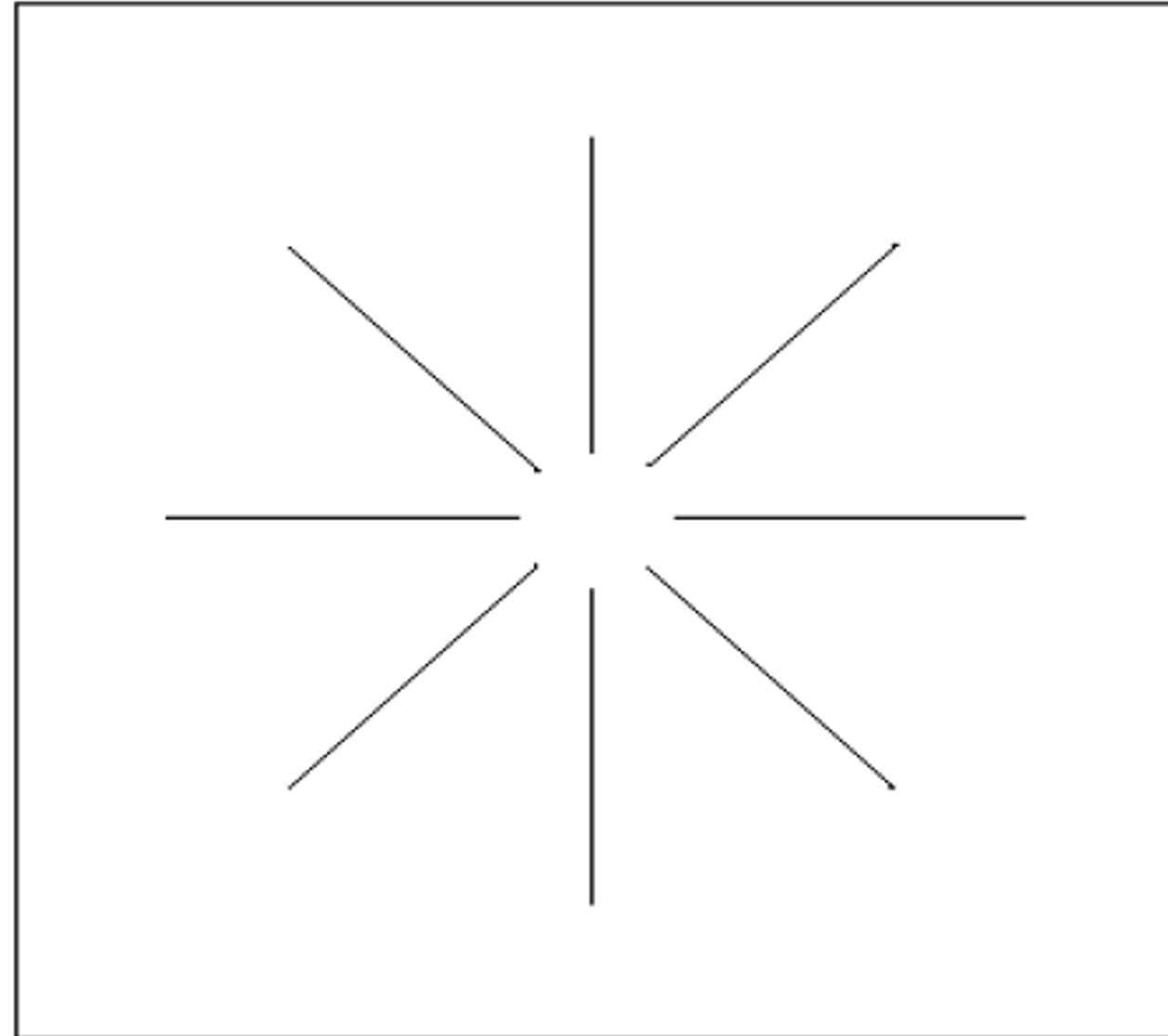
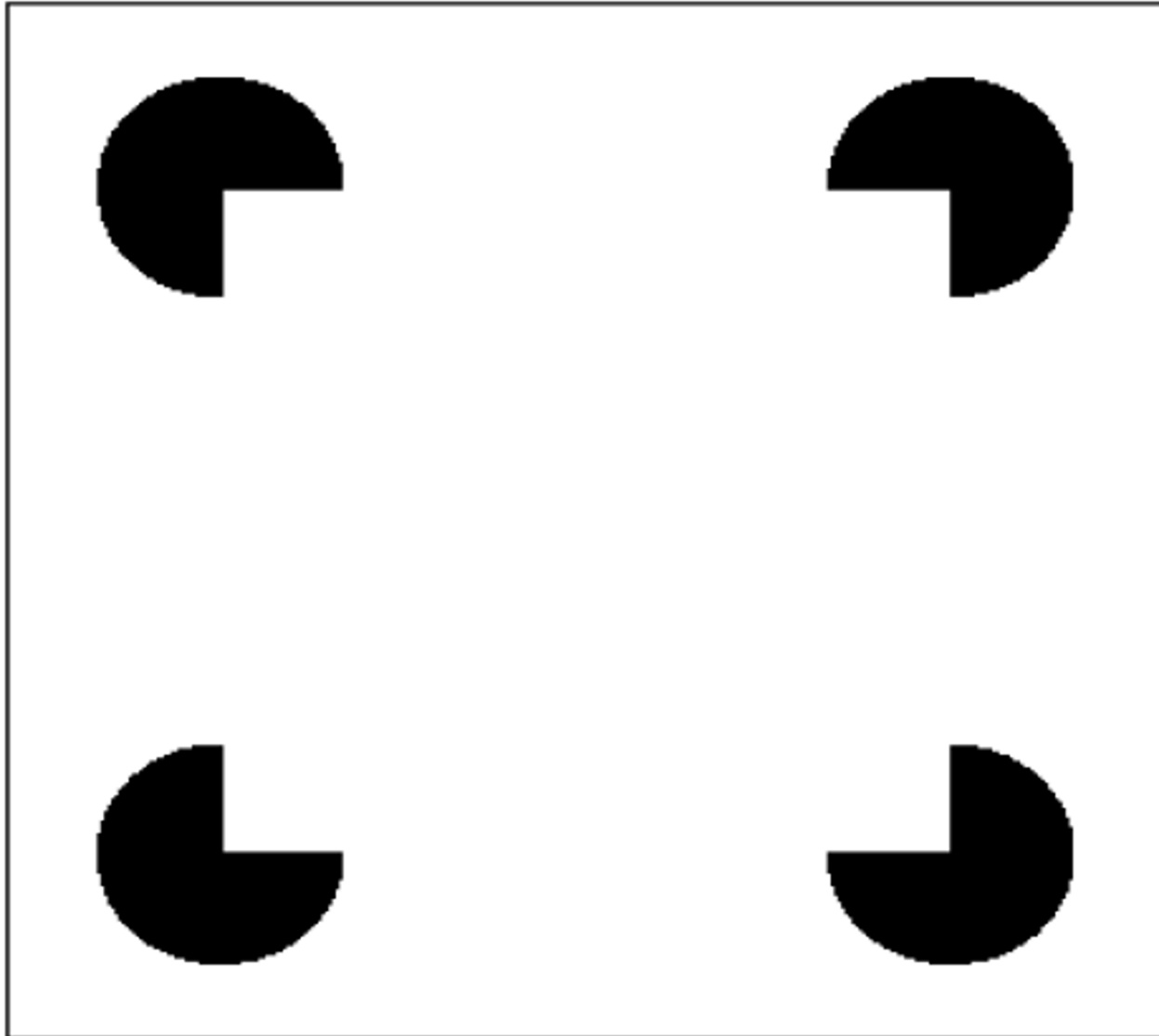


Image Perception: Optical Illusions

- Checkerboard illusion
- The squares marked A and B are ...
- ... the same shade of gray

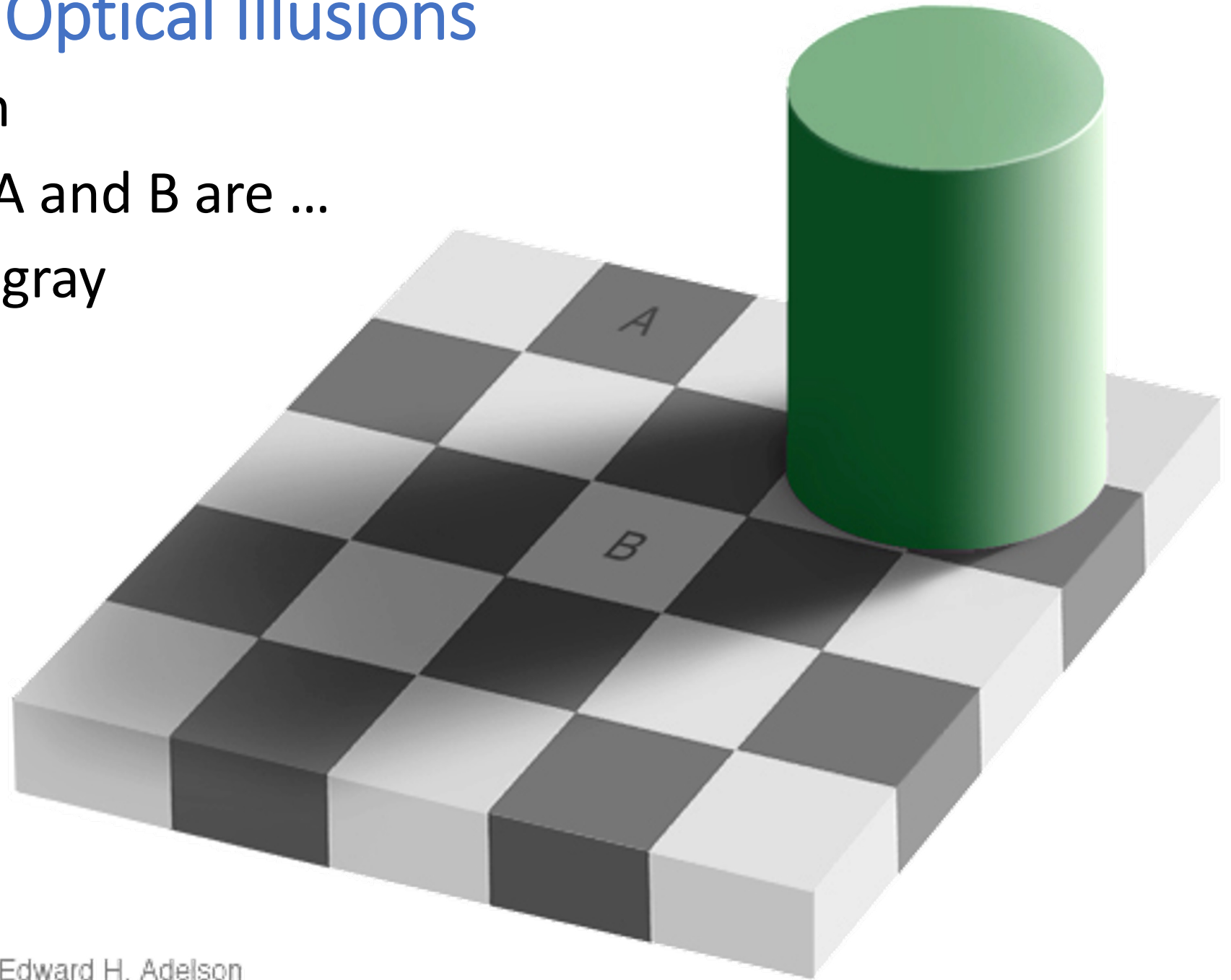


Image Perception: Optical Illusions

- Checkerboard illusion explanations:
 - 1) Simultaneous contrast
 - 2) Shadows
 - 3) Checkerboard pattern

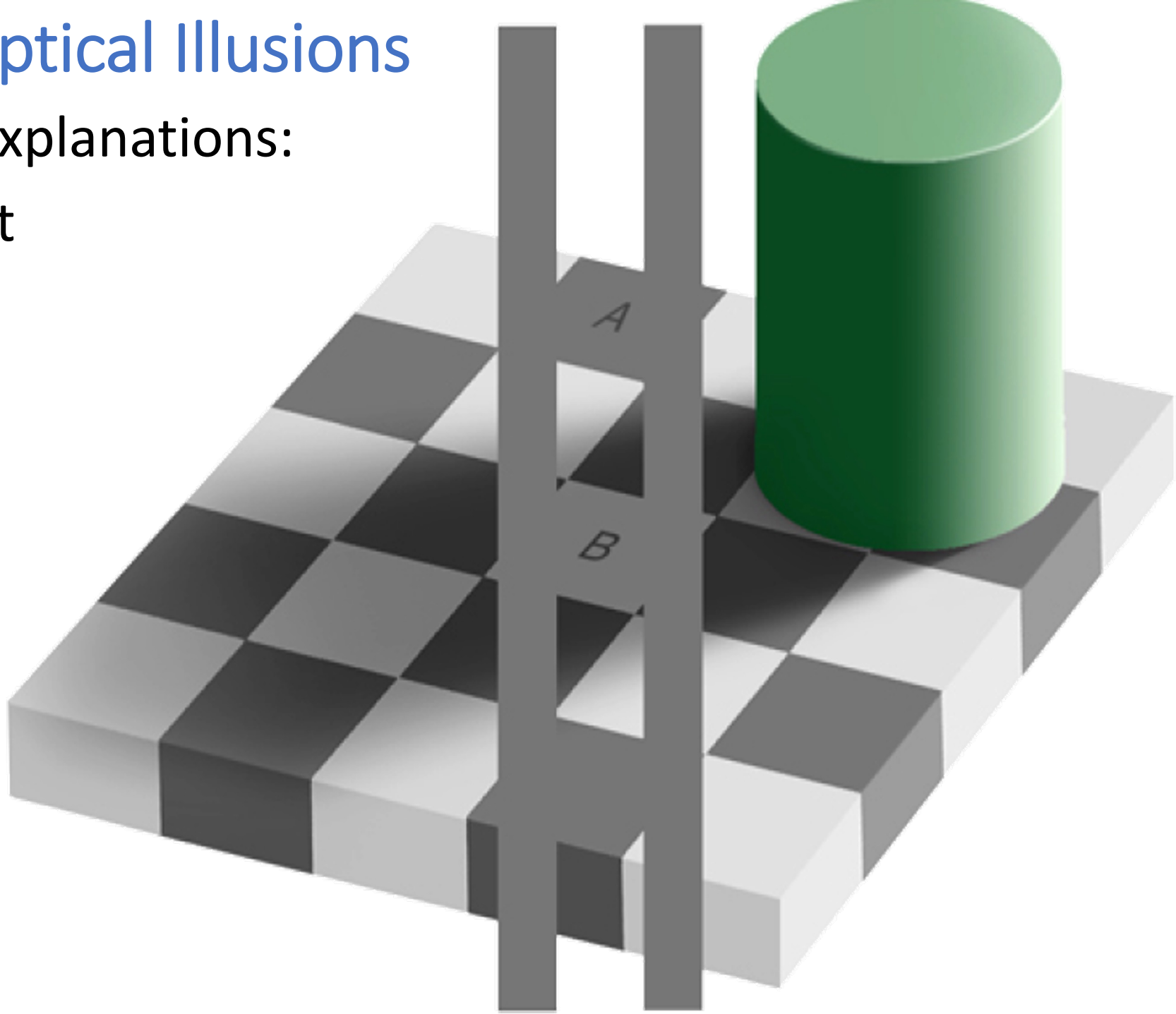


Image Perception: Optical Illusions

- Human visual system
- Is NOT optimized for measuring the amount of light
- Has evolved to achieve high-level understanding from limited data
- e.g., object detection and recognition despite occlusion
- e.g., object detection despite low light

- Be careful while interpreting results of your algorithms !